Published for SISSA by Springer

Received: November 28, 2023
Revised: February 12, 2024
Accepted: February 17, 2024
Published: March 12, 2024

Analytic decay width of the Higgs boson to massive bottom quarks at next-to-next-to-leading order in QCD

Jian Wang,[b,c] **Yefan Wang**[a,b,*] **and Da-Jiang Zhang**[b]

[a] *Department of Physics and Institute of Theoretical Physics, Nanjing Normal University, Nanjing, Jiangsu 210023, China*
[b] *School of Physics, Shandong University, Jinan, Shandong 250100, China*
[c] *Center for High Energy Physics, Peking University, Beijing 100871, China*

E-mail: j.wang@sdu.edu.cn, wangyefan@sdu.edu.cn, zhangdajiang@mail.sdu.edu.cn

Abstract: The Higgs boson decay to a massive bottom quark pair provides the dominant contribution to the Higgs boson width. We present an exact result for such a decay induced by the bottom quark Yukawa coupling with next-to-next-to-leading order (NNLO) QCD corrections. We have adopted the canonical differential equations in the calculation and obtained the result in terms of multiple polylogarithms. We also compute the contribution from the decay to four bottom quarks which consists of complete elliptic integrals or their one-fold integrals. The result in the small bottom quark mass limit coincides with the previous calculation using the large momentum expansion. The threshold expansion exhibits power divergent terms in the bottom quark velocity, which has a structure different from that in $e^+e^- \to t\bar{t}$ but can be reproduced by computing the corresponding Coulomb Green function. The NNLO corrections significantly reduce the uncertainties from both the renormalization scale and the renormalization scheme of the bottom quark Yukawa coupling. Our result can be applied to a heavy scalar decay to a top quark pair.

Keywords: Higgs Properties, Higher-Order Perturbative Calculations, Scattering Amplitudes

ArXiv ePrint: 2310.20514

*Corresponding author.

Open Access, © The Authors.
Article funded by SCOAP[3].

https://doi.org/10.1007/JHEP03(2024)068

JHEP03(2024)068

Contents

1 Introduction **1**

2 Calculation framework **2**

3 Calculation of master integrals **4**
- 3.1 Canonical basis of NP1 5
- 3.2 Canonical basis of P1 8
- 3.3 Boundary conditions 8

4 Renormalization **9**

5 Analytic results **9**
- 5.1 Full results 9
- 5.2 Asymptotic expansion 15

6 Numerical results **19**

7 Conclusion **21**

A Topologies of the master integrals **22**

B Asymptotic expansion of the master integrals **22**

1 Introduction

The Higgs boson was discovered more than ten years ago, and its couplings with other elementary particles in the standard model (SM) have been measured and found to be consistent with the SM predictions [1–3]. The Yukawa couplings of the Higgs boson, along with the spontaneous electroweak symmetry breaking, play a crucial role in explaining the origin of the masses of fermions. The constraints on the Yukawa couplings with the top quark [4, 5], the bottom quark [6, 7], the charm quark [8, 9], the tau lepton [10, 11], and the muon [12, 13] have been set by the ATLAS and CMS collaborations. Because the top quark is very heavy, the Higgs boson can not decay into top quarks. The dominant decay mode of the Higgs boson is thus $H \to b\bar{b}$. From the signal strength observed at the LHC [6, 7], the bottom quark Yukawa coupling is derived with an error of $\pm 10\%$. The accuracy will be significantly improved at future lepton colliders such as the Circular Electron Positron Collider (CEPC) [14–16], and the precision can reach the per mill level [15, 16].

The corresponding theoretical predictions have been improved continuously in the past half-century. The next-to-leading order (NLO) QCD corrections [17–20] were computed more than 40 years ago. And the NLO electroweak (EW) corrections have also been known for a long time [21, 22]. The mixed QCD and EW corrections of order $\mathcal{O}(\alpha\alpha_s)$ have been obtained in refs. [23, 24]. Later, the decay width of $H \to b\bar{b}$ was calculated up to $\mathcal{O}(y_b^2\alpha_s^4)$ [25–29],

JHEP03(2024)068

when the final-state bottom quark is taken to be massless but the bottom Yukawa coupling is still finite. The corresponding differential results were investigated up to $\mathcal{O}(\alpha_s^3)$ [30–32]. The inclusive and differential decay widths with finite m_b were computed at NNLO numerically [33–35]. The effects of parton shower in this decay process have been explored in [36, 37]. The differential decay rates of $H \to b\bar{b}j$ at NNLO and $H \to b\bar{b}b\bar{b}$ at NLO were presented in [38] and [39], respectively.

The decay $H \to b\bar{b}$ can also be induced by a top quark loop with the top quark Yukawa coupling y_t, which is much larger than y_b. However, this contribution starts at the two-loop level, i.e., $\mathcal{O}(y_b y_t \alpha_s^2)$. Because the helicity should be flipped along the bottom quark current, an additional bottom quark mass suppression emerges. As a result, the contribution is not significantly larger, or even smaller, than that of $\mathcal{O}(y_b^2 \alpha_s^2)$. The analytic result of the decay width of $H \to b\bar{b}$ at $\mathcal{O}(y_b y_t \alpha_s^2)$ with finite m_b and m_t has been calculated in [40]. The calculation of the differential decay width that is sensitive to y_t was presented up to $\mathcal{O}(\alpha_s^3)$ in [29, 41]. The decay rate of $H \to$ hadrons including contributions from both $H \to b\bar{b}$ and $H \to gg$ has been computed up to $\mathcal{O}(\alpha_s^4)$ [42, 43].

In this paper, we provide an exact result for the decay $H \to b\bar{b}$ at $\mathcal{O}(y_b^2 \alpha_s^2)$ with full dependence on m_b. Specifically, we consider the contributions from the decay processes $H \to b\bar{b}(+X), X = g, gg, q\bar{q}, b\bar{b}$, where q stands for the massless quark. We note that an analytical calculation has been performed in [44, 45], but only the expansion result in m_b^2/m_H^2 was obtained. Our analytic results enable an efficient computation of the NNLO decay width, and can be readily extended to other similar processes, such as a new heavy scalar decaying to top quarks. Moreover, given the analytic result, we can explore different aspects of the decay rate. The expansion of the decay width near the threshold shows power divergences, which violate the perturbative expansion in α_s. To understand the origin of these divergences and to restore the power of perturbative expansion, we need to calculate a new Coulomb Green function.

This paper is organized as follows. Our calculation framework is introduced in section 2. The master integrals in the calculation of the decay width are computed in section 3. In section 4, the renormalization procedure is briefly described. The analytical and numerical results for the decay width are presented in section 5 and section 6, respectively. Our conclusion is given in section 7.

2 Calculation framework

We are going to calculate the decay width of $H \to b\bar{b}$, denoted by $\Gamma_{Hb\bar{b}}$, via the optical theorem,

$$\Gamma_{Hb\bar{b}} = \frac{\mathrm{Im}(\Sigma)}{m_H}, \tag{2.1}$$

where Σ represents the amplitude of the process $H \to b\bar{b} \to H$. In this method, we do not need to calculate the virtual and real corrections separately, and the complicated multi-body phase space integration can be avoided.

We define the contribution from the final states of two bottom quarks as $\tilde{\Gamma}^{y_b y_b}_{Hb\bar{b}}$, and that from final states of four bottom quarks as $\Gamma^{y_b y_b}_{Hb\bar{b}b\bar{b}}$. The partial decay width of $H \to b\bar{b}$

JHEP03(2024)068

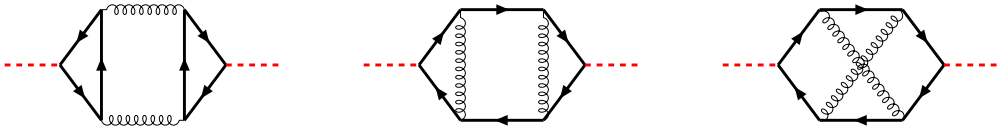

Figure 1. Typical three-loop Feynman diagrams of $H \to b\bar{b} \to H$. The thick black and red lines stand for the massive bottom quark and the Higgs boson, respectively.

Figure 2. The topologies of NP1 and P1 integral families. The thick black and red lines stand for the massive bottom quark and the Higgs boson, respectively. The other lines represent massless particles.

.

is thus given by

$$\Gamma^{y_b y_b}_{Hb\bar{b}} \equiv \tilde{\Gamma}^{y_b y_b}_{Hb\bar{b}} + \Gamma^{y_b y_b}_{Hb\bar{b}b\bar{b}} \,. \tag{2.2}$$

At $\mathcal{O}(y_b^2 \alpha_s^2)$, we have to compute the imaginary part of three-loop self-energy diagrams, as shown in figure 1. We generate the diagrams with the package `FeynArts` [46], and perform the Dirac algebra using the package `FeynCalc` [47]. The amplitudes are expressed as linear combinations of three-loop scalar integrals, which are reduced to a minimal set of integrals called master integrals (MIs) due to the identities from integration by parts (IBP) [48, 49]. The solution of these identities is found by the Laporta algorithm [50] that is implemented in the packages `FIRE` [51] and `Kira` [52]. These MIs can be represented by two integral families, i.e., NP1 and P1 in figure 2. In this procedure the package `CalcLoop`[1] has been used to shift the loop momenta. In our calculation, the Higgs boson and the bottom quark are taken to be massive while the light quarks are massless. We do not consider the top quark contribution.

According to the cutting rules in the optical theorem, the imaginary parts of Σ come from the cut diagrams in which some propagators are put on-shell simultaneously. We find that most cut diagrams in Σ have at least two on-shell bottom-quark propagators. And these cut diagrams correspond to the processes $H \to b\bar{b}$, $H \to b\bar{b}g$, $H \to b\bar{b}gg$, $H \to b\bar{b}q\bar{q}$ and $H \to b\bar{b}b\bar{b}$. All of them contribute to Γ_{Hbb} at $\mathcal{O}(y_b^2 \alpha_s^2)$. Two sample cut Feynman diagrams are shown in figure 3. The requirement of cuts on two bottom quarks simplifies the calculations since some MIs do not have imaginary parts, e.g., the three-loop massive vacuum bubble diagram shown in figure 4.

Notice that some cut diagrams contribute to the imaginary part of Σ, but do not have two bottom-quark cut propagators, such as the diagram of a cut through two gluon propagators as depicted in figure 5. This cut diagram is related to the process $H \to gg$, which should not be included in the corrections of $H \to b\bar{b}$. To subtract this contribution from the imaginary part of the forward scattering amplitude, we calculate the squared amplitude of $H \to gg$ with one triangle bottom-quark loop, and perform the two-body phase space integration.

[1] https://gitlab.com/multiloop-pku/calcloop.

JHEP03(2024)068

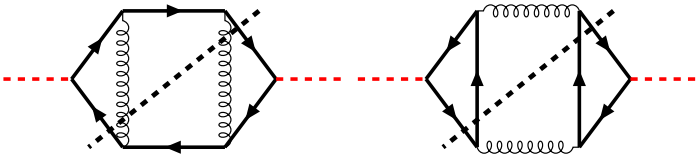

Figure 3. Typical cut diagrams. The black dashed line stands for the cut line, and the propagators crossing the cut line are on-shell. The cut line passes through two (left graph) and four (right graph) bottom quarks.

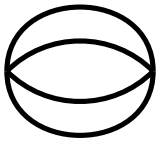

Figure 4. A three-loop massive vacuum bubble diagram.

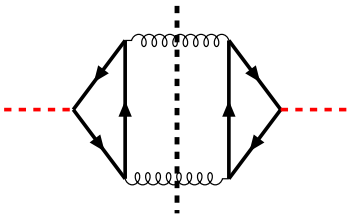

Figure 5. A cut diagram contributing to the $H \to gg$ decay.

Figure 6. Two master integrals contributing to the decay to four bottom-quark final states. The thick black and red lines stand for the massive bottom quark and the Higgs boson, respectively.

3 Calculation of master integrals

As mentioned in eq. (2.2), we have to calculate the contributions from both two and four bottom quark final states. The corresponding MIs can be computed separately.

The MIs contributing to the decay to four bottom quark final state contain elliptic integrals, such as those shown in figure 6. They have been calculated in [53] where the authors consider the total cross section of $e^+e^- \to Q\bar{Q}Q\bar{Q}$. After choosing a regular basis, only the $\mathcal{O}(\epsilon^0)$ parts of the MIs are needed, which in turn can be expressed either as complete elliptic integrals of the first kind or one-fold integrals of them. Following this method, we present the decay width $\Gamma^{y_b y_b}_{Hb\bar{b}b\bar{b}}$ in terms of one-fold integrals over expressions depending on complete elliptic integrals and multiple polylogarithms. The explicit form can be found in eq. (5.8) below.

Then we turn to the MIs that are needed in the calculation of $\tilde{\Gamma}^{y_b y_b}_{Hb\bar{b}}$. We adopt the method of differential equations [54, 55]. Once a set of canonical basis $\boldsymbol{F}$ is found, the

JHEP03(2024)068

Figure 7. The four (left) and two (right) bottom quark cut diagrams. The thick black and red lines stand for the massive bottom quark and the Higgs boson, respectively.

dimensional regulator ϵ is factorized out from the kinematic variables [56],

$$d\boldsymbol{F}(\epsilon, x_n) = \epsilon\, dA(x_n)\boldsymbol{F}(\epsilon, x_n), \tag{3.1}$$

where $\{x_n\}$ denote the kinematic variables. When $dA(x_n)$ is expressed in $d\log$ forms, the differential equations can be solved recursively in terms of multiple polylogarithms (MPLs) [57], which are defined by $G(x) \equiv 1$ and

$$G(l_1, l_2, \ldots, l_n, x) \equiv \int_0^x \frac{\mathrm{d}t}{t - l_1} G(l_2, \ldots, l_n, t)\,, \tag{3.2}$$

$$G(\overrightarrow{0}_n, x) \equiv \frac{1}{n!}\ln^n x\,. \tag{3.3}$$

The number of elements in the set $\{l_1, l_2, \ldots, l_n\}$ is referred to as the transcendental *weight* of the MPLs. Note that we have dropped MIs that possess an imaginary part only from cuts on four bottom quarks. And all the boundary values are evaluated in the phase space where it is not possible to apply a cut on four bottom quarks. For instance, the left diagram in figure 6 is dropped in the basis. The right diagram in figure 6 is included but only the cut through two bottom quarks, as shown in the right diagram of figure 7, is considered.

3.1 Canonical basis of NP1

The master integrals of the NP1 integral family in figure 2 are defined by

$$I^{\mathrm{NP1}}_{n_1,n_2,\ldots,n_9} = \mathrm{Im}\int \mathcal{D}^D q_1\, \mathcal{D}^D q_2 \frac{D_9^{-n_9}}{D_1^{n_1}\, D_2^{n_2}\, D_3^{n_3}\, D_4^{n_4}\, D_5^{n_5}\, D_6^{n_6}\, D_7^{n_7} D_8^{n_8}} \tag{3.4}$$

with

$$\mathcal{D}^D q_i = \frac{(m_b^2)^\epsilon}{i\pi^{D/2}\Gamma(1+\epsilon)} d^D q_i\,, \quad D = 4 - 2\epsilon\,, \tag{3.5}$$

where all $n_i \geq 0, i = 1, \cdots, 8$, and $n_9 \leq 0$. The denominators D_i read

$$\begin{aligned}
&D_1 = q_1^2 - m_b^2, && D_2 = q_2^2 - m_b^2, && D_3 = (q_1 - q_2)^2,\\
&D_4 = q_3^2 - m_b^2, && D_5 = (q_2 + q_3)^2, && D_6 = (q_3 - k)^2 - m_b^2,\\
&D_7 = (q_1 + k)^2 - m_b^2, && D_8 = (q_1 - q_2 - q_3 + k)^2 - m_b^2, && D_9 = (q_1 - q_3 + k)^2,
\end{aligned} \tag{3.6}$$

where the Feynman prescription $+i\varepsilon$ has been suppressed. The external momentum k satisfies $k^2 = m_H^2$. We focus only on the imaginary part of the integral, which gets contributions from all possible cuts except for the ones on four bottom quarks. The results of the master integrals for specific cuts have been obtained in [59].

JHEP03(2024)068

In the NP1 integral family, there are 29 MIs contributing to $\tilde{\Gamma}_{Hb\bar{b}}$. To construct the canonical basis, we first select the MIs

$$
\begin{aligned}
&\mathrm{M}_1^{\mathrm{NP1}} = \epsilon^3 m_b^2 I^{\mathrm{NP1}}_{2,2,0,2,0,1,0,0,0}, && \mathrm{M}_2^{\mathrm{NP1}} = \epsilon^3 m_b^2 I^{\mathrm{NP1}}_{2,2,0,0,1,2,0,0,0}, \\
&\mathrm{M}_3^{\mathrm{NP1}} = \epsilon^3 m_b^2 I^{\mathrm{NP1}}_{2,2,0,0,2,1,0,0,0}, && \mathrm{M}_4^{\mathrm{NP1}} = \epsilon^3 m_b^2 I^{\mathrm{NP1}}_{2,0,2,0,1,2,0,0,0}, \\
&\mathrm{M}_5^{\mathrm{NP1}} = \epsilon^3 m_b^2 I^{\mathrm{NP1}}_{2,0,2,0,2,1,0,0,0}, && \mathrm{M}_6^{\mathrm{NP1}} = (1-2\epsilon)\epsilon m_b^4 I^{\mathrm{NP1}}_{3,0,1,0,1,3,0,0,0}, \\
&\mathrm{M}_7^{\mathrm{NP1}} = \epsilon^3 m_b^4 I^{\mathrm{NP1}}_{2,2,0,2,0,1,1,0,0}, && \mathrm{M}_8^{\mathrm{NP1}} = \epsilon^3 m_b^4 I^{\mathrm{NP1}}_{0,2,1,2,0,1,2,0,0}, \\
&\mathrm{M}_9^{\mathrm{NP1}} = \epsilon^3 m_b^4 I^{\mathrm{NP1}}_{0,2,2,2,0,1,1,0,0}, && \mathrm{M}_{10}^{\mathrm{NP1}} = \epsilon^3 m_b^2 I^{\mathrm{NP1}}_{0,1,1,0,1,2,2,0,0}, \\
&\mathrm{M}_{11}^{\mathrm{NP1}} = \epsilon^2 m_b^4 I^{\mathrm{NP1}}_{0,1,1,0,1,3,2,0,0}, && \mathrm{M}_{12}^{\mathrm{NP1}} = (1-2\epsilon)\epsilon^3 m_b^2 I^{\mathrm{NP1}}_{0,2,1,0,1,2,1,0,0}, \\
&\mathrm{M}_{13}^{\mathrm{NP1}} = (1-2\epsilon)\epsilon^3 m_b^2 I^{\mathrm{NP1}}_{2,2,0,0,1,1,0,1,0}, && \mathrm{M}_{14}^{\mathrm{NP1}} = \epsilon^3 m_b^4 I^{\mathrm{NP1}}_{2,2,0,0,1,2,0,1,0}, \\
&\mathrm{M}_{15}^{\mathrm{NP1}} = (1-2\epsilon)\epsilon^3 m_b^2 I^{\mathrm{NP1}}_{2,0,2,1,1,1,0,0,0}, && \mathrm{M}_{16}^{\mathrm{NP1}} = (1-2\epsilon)\epsilon^3 m_b^2 I^{\mathrm{NP1}}_{2,2,0,1,0,1,0,1,0}, \\
&\mathrm{M}_{17}^{\mathrm{NP1}} = (1-2\epsilon)\epsilon^4 m_b^2 I^{\mathrm{NP1}}_{1,0,2,1,1,1,1,0,0}, && \mathrm{M}_{18}^{\mathrm{NP1}} = (1-2\epsilon)\epsilon^4 m_b^2 I^{\mathrm{NP1}}_{1,1,1,0,1,2,1,0,0}, \\
&\mathrm{M}_{19}^{\mathrm{NP1}} = (1-2\epsilon)\epsilon^4 m_b^2 I^{\mathrm{NP1}}_{2,1,0,1,1,1,0,1,0}, && \mathrm{M}_{20}^{\mathrm{NP1}} = (1-2\epsilon)\epsilon^3 m_b^4 I^{\mathrm{NP1}}_{2,1,0,1,1,1,0,2,0}, \\
&\mathrm{M}_{21}^{\mathrm{NP1}} = (1-2\epsilon)\epsilon^5 m_b^2 I^{\mathrm{NP1}}_{1,1,1,1,1,1,0,1,0}, && \mathrm{M}_{22}^{\mathrm{NP1}} = (1-2\epsilon)^2\epsilon^4 m_b^2 I^{\mathrm{NP1}}_{1,1,1,0,0,1,1,1,1}, \\
&\mathrm{M}_{23}^{\mathrm{NP1}} = (1-2\epsilon)\epsilon^4 m_b^4 I^{\mathrm{NP1}}_{1,1,1,0,0,1,2,1,1}, && \mathrm{M}_{24}^{\mathrm{NP1}} = (1-2\epsilon)\epsilon^4 m_b^2 I^{\mathrm{NP1}}_{1,2,0,1,0,1,1,1,0}, \\
&\mathrm{M}_{25}^{\mathrm{NP1}} = (1-2\epsilon)\epsilon^3 m_b^4 I^{\mathrm{NP1}}_{2,2,0,1,0,1,1,1,0}, && \mathrm{M}_{26}^{\mathrm{NP1}} = (1-2\epsilon)\epsilon^4 m_b^4 I^{\mathrm{NP1}}_{1,2,1,1,1,1,1,0,0}, \\
&\mathrm{M}_{27}^{\mathrm{NP1}} = (1-2\epsilon)\epsilon^4 m_b^4 I^{\mathrm{NP1}}_{2,1,1,1,1,1,1,0,0}, && \mathrm{M}_{28}^{\mathrm{NP1}} = (1-2\epsilon)\epsilon^5 m_b^4 I^{\mathrm{NP1}}_{1,1,1,1,1,1,1,1,0}, \\
&\mathrm{M}_{29}^{\mathrm{NP1}} = (1-2\epsilon)\epsilon^5 m_b^2 I^{\mathrm{NP1}}_{1,1,1,1,1,1,1,1,-1}\,.
\end{aligned}
\tag{3.7}
$$

Here all $\mathrm{M}_i^{\mathrm{NP1}}$ are dimensionless, and the corresponding topology diagrams are displayed in figure 11 in appendix A.

Then the canonical basis integrals F_i^{NP1}, $i = 1, \ldots, 29$, can be constructed as linear combinations of $\mathrm{M}_i^{\mathrm{NP1}}$ using a method similar to that in ref. [58]. Examples can be found in [60]. For simplicity, we define a dimensionless variable $z \equiv m_H^2/m_b^2 + i0^+$. Then we obtain the following canonical basis of the NP1 family:

$$
F_1^{\mathrm{NP1}} = \mathrm{M}_1^{\mathrm{NP1}} r, \qquad F_2^{\mathrm{NP1}} = \mathrm{M}_2^{\mathrm{NP1}}(-z), \qquad F_3^{\mathrm{NP1}} = \frac{\mathrm{M}_2^{\mathrm{NP1}} r}{2} + \mathrm{M}_3^{\mathrm{NP1}} r,
$$

$$
F_4^{\mathrm{NP1}} = \mathrm{M}_4^{\mathrm{NP1}}(-z), \qquad F_5^{\mathrm{NP1}} = \left(\mathrm{M}_4^{\mathrm{NP1}} + \mathrm{M}_5^{\mathrm{NP1}}\right)(-r), \qquad F_6^{\mathrm{NP1}} = -\frac{\mathrm{M}_5^{\mathrm{NP1}}(4-z)}{4} + \mathrm{M}_6^{\mathrm{NP1}},
$$

$$
F_7^{\mathrm{NP1}} = \mathrm{M}_7^{\mathrm{NP1}} z(z-4), \qquad F_8^{\mathrm{NP1}} = \mathrm{M}_8^{\mathrm{NP1}} r z, \qquad F_9^{\mathrm{NP1}} = \frac{\mathrm{M}_8^{\mathrm{NP1}}(z-4)z}{2} + \mathrm{M}_9^{\mathrm{NP1}}(z-4)z,
$$

$$
\begin{aligned}
F_{10}^{\mathrm{NP1}} = {} & \frac{\mathrm{M}_2^{\mathrm{NP1}}(4-3z)}{8-4z} + \frac{\mathrm{M}_4^{\mathrm{NP1}} z}{4} + \frac{\mathrm{M}_{10}^{\mathrm{NP1}}\left(2\epsilon\left(4z^2-9z-4\right)+4z^2-11z+4\right)}{8-4z} \\
& - \frac{2\mathrm{M}_{11}^{\mathrm{NP1}}\epsilon(2z-5)z}{z-2} + \frac{\mathrm{M}_{12}^{\mathrm{NP1}}(z-4)(z-1)}{4(z-2)},
\end{aligned}
$$

$$
\begin{aligned}
F_{11}^{\mathrm{NP1}} = {} & \frac{\mathrm{M}_2^{\mathrm{NP1}} r\left(z^2-6z+4\right)}{8(z-4)(z-2)} + \frac{\mathrm{M}_{10}^{\mathrm{NP1}} r\left(2\epsilon\left(z^2+2z-11\right)+z^2-4z+3\right)}{2(z-4)(z-2)} \\
& + \frac{\mathrm{M}_{11}^{\mathrm{NP1}} r\left(2\epsilon\left(z^2-6\right)-z+2\right)}{(z-4)(z-2)} - \frac{\mathrm{M}_{12}^{\mathrm{NP1}} r(z-3)(z-1)}{2(z-4)(z-2)},
\end{aligned}
$$

JHEP03(2024)068

$$
\begin{aligned}
&F_{12}^{\mathrm{NP1}}=-\frac{\mathrm{M}_2^{\mathrm{NP1}}z}{z-2}+\mathrm{M}_4^{\mathrm{NP1}}z+\frac{\mathrm{M}_{10}^{\mathrm{NP1}}(6\epsilon z+z)}{2-z}-\frac{8\mathrm{M}_{11}^{\mathrm{NP1}}\epsilon z}{z-2}+\frac{\mathrm{M}_{12}^{\mathrm{NP1}}(z-1)z}{z-2},\\
&F_{13}^{\mathrm{NP1}}=\mathrm{M}_{13}^{\mathrm{NP1}}z+\mathrm{M}_{14}^{\mathrm{NP1}}z^2,\qquad F_{14}^{\mathrm{NP1}}=\mathrm{M}_{14}^{\mathrm{NP1}}(-z),\\
&F_{15}^{\mathrm{NP1}}=-\frac{\mathrm{M}_4^{\mathrm{NP1}}r(7z-12)}{12(4-z)}+\frac{\mathrm{M}_5^{\mathrm{NP1}}r}{12}-\frac{\mathrm{M}_6^{\mathrm{NP1}}r}{3(4-z)}+\frac{\mathrm{M}_{15}^{\mathrm{NP1}}r}{4-z},\qquad F_{16}^{\mathrm{NP1}}=-\frac{\mathrm{M}_{16}^{\mathrm{NP1}}r}{4-z},\\
&F_{17}^{\mathrm{NP1}}=\mathrm{M}_{17}^{\mathrm{NP1}}(-z),\qquad F_{18}^{\mathrm{NP1}}=\mathrm{M}_{18}^{\mathrm{NP1}}(-z),\qquad F_{19}^{\mathrm{NP1}}=\mathrm{M}_{19}^{\mathrm{NP1}}(-z),\\
&F_{20}^{\mathrm{NP1}}=-2\mathrm{M}_{14}^{\mathrm{NP1}}rz-\frac{3\mathrm{M}_{16}^{\mathrm{NP1}}rz}{z-4}+\mathrm{M}_{20}^{\mathrm{NP1}}rz,\qquad F_{21}^{\mathrm{NP1}}=\mathrm{M}_{21}^{\mathrm{NP1}}z,\\
&F_{22}^{\mathrm{NP1}}=\frac{1}{4}z\left(\mathrm{M}_{22}^{\mathrm{NP1}}+\mathrm{M}_{23}^{\mathrm{NP1}}(z-4)\right),\\
&F_{23}^{\mathrm{NP1}}=-\mathrm{M}_{18}^{\mathrm{NP1}}r+2\mathrm{M}_{19}^{\mathrm{NP1}}r+2\mathrm{M}_{21}^{\mathrm{NP1}}r+\mathrm{M}_{22}^{\mathrm{NP1}}r+\mathrm{M}_{23}^{\mathrm{NP1}}rz,\\
&F_{24}^{\mathrm{NP1}}=\mathrm{M}_{24}^{\mathrm{NP1}}(-z),\qquad F_{25}^{\mathrm{NP1}}=-\frac{\mathrm{M}_{16}^{\mathrm{NP1}}rz}{z-4}+\mathrm{M}_{25}^{\mathrm{NP1}}rz,\qquad F_{26}^{\mathrm{NP1}}=\mathrm{M}_{26}^{\mathrm{NP1}}(-z),\\
&F_{27}^{\mathrm{NP1}}=\frac{3\mathrm{M}_2^{\mathrm{NP1}}rz^2}{8(z-2)^2}+\frac{3\mathrm{M}_4^{\mathrm{NP1}}rz}{8(z-2)}-\frac{3\mathrm{M}_8^{\mathrm{NP1}}r(z-4)z}{8(z-2)}-\frac{3\mathrm{M}_9^{\mathrm{NP1}}r(z-4)z}{4(z-2)}\\
&\qquad+\frac{3\mathrm{M}_{10}^{\mathrm{NP1}}r(6\epsilon+1)z}{4(z-2)^2}+\frac{6\mathrm{M}_{11}^{\mathrm{NP1}}r\epsilon z}{(z-2)^2}-\frac{3\mathrm{M}_{12}^{\mathrm{NP1}}r(z-1)z}{4(z-2)^2}+\frac{3\mathrm{M}_{17}^{\mathrm{NP1}}rz}{2(z-2)}\\
&\qquad+\frac{\mathrm{M}_{18}^{\mathrm{NP1}}rz}{4(z-2)}+\frac{\mathrm{M}_{26}^{\mathrm{NP1}}rz}{z-2}-\frac{\mathrm{M}_{27}^{\mathrm{NP1}}rz}{z-2},\\
&F_{28}^{\mathrm{NP1}}=\mathrm{M}_{28}^{\mathrm{NP1}}(-r)z,\qquad F_{29}^{\mathrm{NP1}}=\mathrm{M}_{28}^{\mathrm{NP1}}(z-4)z+\mathrm{M}_{29}^{\mathrm{NP1}}z,
\end{aligned}
\tag{3.8}
$$

where

$$
r=\sqrt{z(z-4)}. \tag{3.9}
$$

To rationalize the square root r, we write

$$
z=-\frac{(w-1)^2}{w} \tag{3.10}
$$

with $-1<w<0$. Then r is rationalized to

$$
r=\frac{(w+1)(w-1)}{w}. \tag{3.11}
$$

The corresponding differential equations for the canonical basis $\boldsymbol{F}^{\mathrm{NP1}}$ become

$$
\frac{d\boldsymbol{F}^{\mathrm{NP1}}}{dw}=\epsilon\left(\sum_{i=1}^{5}\frac{\boldsymbol{N}_{\boldsymbol{i}}^{\mathrm{NP1}}}{w-l_i^{\mathrm{NP1}}}\right)\boldsymbol{F}^{\mathrm{NP1}} \tag{3.12}
$$

with

$$
l_1^{\mathrm{NP1}}=0\,,\ l_2^{\mathrm{NP1}}=1\,,\ l_3^{\mathrm{NP1}}=-1\,,\ l_4^{\mathrm{NP1}}=\frac{1+\sqrt{3}i}{2}\,,\ l_5^{\mathrm{NP1}}=\frac{1-\sqrt{3}i}{2} \tag{3.13}
$$

and $\boldsymbol{N}_{\boldsymbol{i}}^{\mathrm{NP1}}$ being constant rational matrices.

JHEP03(2024)068

3.2 Canonical basis of P1

The P1 integral family is defined by

$$I^{P1}_{n_1,n_2,\ldots,n_9} = \mathrm{Im}\int \mathcal{D}^D q_1\, \mathcal{D}^D q_2 \frac{D_9^{-n_9}}{D_1^{n_1}\, D_2^{n_2}\, D_3^{n_3}\, D_4^{n_4}\, D_5^{n_5}\, D_6^{n_6}\, D_7^{n_7} D_8^{n_8}}, \tag{3.14}$$

where the denominators D_i read

$$\begin{aligned} &D_1 = q_1^2 - m_b^2, && D_2 = (q_1+k_1)^2 - m_b^2, && D_3 = q_3^2 - m_b^2, \\ &D_4 = (q_3^2 - k_1)^2 - m_b^2, && D_5 = (q_1 - q_2)^2 - m_b^2, && D_6 = q_2^2, \\ &D_7 = (q_2+q_3)^2 - m_b^2, && D_8 = (q_2+k_1)^2, && D_9 = (q_1+q_3)^2. \end{aligned} \tag{3.15}$$

There are 19 MIs in this integral family contributing to $\tilde{\Gamma}_{Hb\bar{b}}$. Most of them have appeared in the NP1 integral family discussed above. Here we only show the new MIs,

$$\begin{aligned} &\mathrm{M}^{\mathrm{P1}}_1 = \epsilon^3 m_b^2 I^{\mathrm{P1}}_{2,0,2,0,0,2,0,1,0}, && \mathrm{M}^{\mathrm{P1}}_5 = \epsilon^3 m_b^4 I^{\mathrm{P1}}_{2,1,2,0,0,2,0,1,0}, \\ &\mathrm{M}^{\mathrm{P1}}_{12} = \epsilon^3 m_b^6 I^{\mathrm{P1}}_{2,1,2,1,0,2,0,1,0}, && \mathrm{M}^{\mathrm{P1}}_{13} = (1-2\epsilon)\epsilon^4 m_b^2 I^{\mathrm{P1}}_{1,1,2,0,1,1,0,1,0}, \\ &\mathrm{M}^{\mathrm{P1}}_{18} = (1-2\epsilon)\epsilon^4 m_b^4 I^{\mathrm{P1}}_{1,1,2,1,1,1,0,1,0}, && \mathrm{M}^{\mathrm{P1}}_{19} = (1-2\epsilon)\epsilon^5 m_b^4 I^{\mathrm{P1}}_{1,1,1,1,1,1,1,1,0}. \end{aligned} \tag{3.16}$$

The topology diagrams are displayed in figure 12 in appendix A. We find the corresponding canonical basis integrals,

$$\begin{aligned} &F^{\mathrm{P1}}_1 = \mathrm{M}^{\mathrm{P1}}_1(-z), \quad F^{\mathrm{P1}}_5 = \mathrm{M}^{\mathrm{P1}}_5(-r)z, \quad F^{\mathrm{P1}}_{12} = -\mathrm{M}^{\mathrm{P1}}_{12} z^2 (z-4), \\ &F^{\mathrm{P1}}_{13} = \mathrm{M}^{\mathrm{P1}}_{13}(-z), \quad F^{\mathrm{P1}}_{18} = \mathrm{M}^{\mathrm{P1}}_{18}(-r)z, \quad F^{\mathrm{P1}}_{19} = \mathrm{M}^{\mathrm{P1}}_{19} z^2. \end{aligned} \tag{3.17}$$

The corresponding differential equations are

$$\frac{d\boldsymbol{F}^{\mathrm{P1}}}{dw} = \epsilon \left(\sum_{i=1}^{3} \frac{\boldsymbol{N}_i^{\mathrm{P1}}}{w - l_i^{\mathrm{P1}}} \right) \boldsymbol{F}^{\mathrm{P1}} \tag{3.18}$$

with

$$l_1^{\mathrm{P1}} = 0\,,\ l_2^{\mathrm{P1}} = 1\,,\ l_3^{\mathrm{P1}} = -1\,, \tag{3.19}$$

and $\boldsymbol{N}_i^{\mathrm{P1}}$ being constant rational matrices.

3.3 Boundary conditions

Given the above canonical differential equations, we obtain the results of the basis integrals at each order of ϵ up to some integration constants. The MPLs in the results can be evaluated efficiently using the `PolyLogTools` package [61], which depends on the algorithm of ref. [62] in the `GiNaC` [63] framework. The integration constants are calculated numerically by comparing the results of basis integrals with those obtained using the `AMFlow` package [64–67]. Their analytical form is reconstructed using the PSLQ algorithm [68, 69]. We find that the transcendental constants relevant to our calculation consist of

$$\{\log(2), \pi, \zeta(3)\} \tag{3.20}$$

up to weight four, where $\zeta(n)$ is the Riemann zeta function. All the master integrals are then checked against the numerical computation with `AMFlow`, and over 200-digit agreement is achieved. We provide the analytical results of the canonical bases in the supplementary material attached to this paper.

JHEP03(2024)068

4 Renormalization

In the framework of the optical theorem, the imaginary part of each three-loop self energy diagram arises from all possible cuts, corresponding to the sum of both virtual and real corrections. The infrared divergences cancel out in the sum, and only ultraviolet divergences remain.

We perform renormalization following the standard procedure, i.e., by including the contribution from diagrams with counter-terms. The bottom quark mass is renormalized in the on-shell scheme, while the strong coupling α_s is renormalized in the $\overline{\text{MS}}$ scheme,

$$\begin{aligned} m_{b,\text{bare}} &= Z_m m_b, \\ \alpha_s^{\text{bare}} C^\epsilon &= \mu^{2\epsilon} \alpha_s Z_{\alpha_s}, \end{aligned} \tag{4.1}$$

where $C_\epsilon = (4\pi e^{-\gamma_E})^\epsilon$. The renormalization constants up to α_s^2 are given by [70–73],

$$\begin{aligned} Z_{\alpha_s} &= 1 + \left(\frac{\alpha_s}{4\pi}\right) \frac{2n_l - 31}{3\epsilon} + \left(\frac{\alpha_s}{4\pi}\right)^2 \left(\frac{(31-2n_l)^2}{9\epsilon^2} + \frac{19n_l - 134}{3\epsilon}\right), \\ Z_m^{\overline{\text{MS}}} &= 1 - \left(\frac{\alpha_s}{4\pi}\right) \frac{4}{\epsilon} + \left(\frac{\alpha_s}{4\pi}\right)^2 \left(\frac{2(43-2n_l)}{3\epsilon^2} + \frac{10n_l - 293}{9\epsilon}\right), \\ Z_m^{\text{OS}} &= 1 + \left(\frac{\alpha_s}{4\pi}\right) D_\epsilon \left(-\frac{4}{\epsilon} - \frac{16}{3} - \frac{32\epsilon}{3} - \frac{64\epsilon^2}{3}\right) \\ &\quad + \left(\frac{\alpha_s}{4\pi}\right)^2 D_\epsilon^2 \Bigg(\frac{86 - 4n_l}{3\epsilon^2} + \frac{10n_l - 101}{9\epsilon} + \frac{4\,(2n_l - 31)}{3\epsilon} \log\left(\frac{\mu^2}{m^2}\right) \\ &\quad + \frac{2(31-2n_l)}{3} \log^2\left(\frac{\mu^2}{m^2}\right) + \frac{16(2n_l - 31)}{9} \log\left(\frac{\mu^2}{m^2}\right) + \frac{(10n_l - 79)\,\pi^2}{9} \\ &\quad + \frac{(142n_l - 2251)}{18} + \frac{8\zeta(3)}{3} - \frac{16\pi^2 \log(2)}{9}\Bigg), \end{aligned} \tag{4.2}$$

where

$$D_\epsilon \equiv \frac{\Gamma(1+\epsilon)}{e^{-\gamma_E \epsilon}} \left(\frac{\mu^2}{m^2}\right)^\epsilon \tag{4.3}$$

and color factors $C_F = 4/3$, $C_A = 3$, $T_R = 1/2$ have been substituted.

The bottom quark Yukawa coupling is related to the bottom quark mass. However, one has the freedom to choose a different renormalization scheme. Though it is expected that the final result for the decay width should be the same if all order corrections are included, the fixed-order results in different schemes may exhibit obvious deviations, which should be taken as one kind of theoretical uncertainty. The results with Yukawa couplings in the on-shell and $\overline{\text{MS}}$ scheme can be converted to each other by applying the relations between the two schemes, which have been computed up to $\mathcal{O}(\alpha_s^4)$ [74–77]. We have performed two separate calculations using different renormalization schemes. The ultraviolet divergences cancel in either scheme, and the final results agree with each other after the conversion of the renormalization scheme.

5 Analytic results

5.1 Full results

The decay width of $H \to b\bar{b}$ up to $\mathcal{O}(\alpha_s^2)$ can be expressed as

$$\Gamma(H \to b\bar{b}) = \Gamma_{\text{LO}} \left[1 + \frac{\alpha_s(\mu)}{\pi} X_1^{y_b y_b} + \left(\frac{\alpha_s(\mu)}{\pi}\right)^2 (X_2^{y_b y_b} + X_2^{y_b y_t})\right]. \tag{5.1}$$

JHEP03(2024)068

The LO result is

$$\Gamma_{\rm LO} = \frac{N_c}{16\pi}\overline{y}_b^2 m_H \left(1-\frac{4}{z}\right)^{3/2}, \tag{5.2}$$

where $N_c = 3$ is the number of colors and $\overline{y}_b$ is the $\overline{\rm MS}$ Yukawa coupling

$$\overline{y}_b(\mu) = \overline{m_b}(\mu)\left(2\sqrt{2}G_F\right)^{1/2} \tag{5.3}$$

with $\overline{m_b}$ the $\overline{\rm MS}$ mass of the bottom quark and G_F the Fermi constant.

The NLO coefficient $X_1^{y_b y_b}$ reads

$$\begin{aligned}
X_1^{y_b y_b}(w) = {} & 2\log\left(\frac{\mu^2}{m_H^2}\right) + \frac{8}{3(w^2-1)}\Big[2\left(-w^2+i\pi\left(w^2+1\right)+1\right)G(-1,w) \\
& + \frac{1}{2}\left(w^2+2i\pi\left(w^2+1\right)-1\right)G(1,w) + \frac{w\left(3w^3-w^2-8w-7\right)G(0,w)}{2(w+1)^2} \\
& + \left(w^2+1\right)\left(-2G(-1,0,w)+2G(0,-1,w)+G(0,1,w)-G(1,0,w)\right) \\
& + \frac{(17-12i\pi)w^4+(40+4i\pi)w^3+32i\pi w^2+(-40+28i\pi)w-17}{8(w+1)^2}\Big].
\end{aligned} \tag{5.4}$$

Though there are $i\pi$ terms in the expression, the full result is real since the MPLs also contain imaginary parts.

The $\mathcal{O}(\alpha_s^2)$ corrections include two parts, i.e., $X_2^{y_b y_b}$ and $X_2^{y_b y_t}$. The latter denotes the contribution from the interference of the bottom and top quark Yukawa coupling induced amplitudes and starts at $\mathcal{O}(\alpha_s^2)$. The corresponding analytical result has been calculated in ref. [40]. The former receives contributions from both two and four bottom quark final states, as discussed in section 2,

$$X_2^{y_b y_b} = \tilde{X}_{2,b\bar{b}}^{y_b y_b} + X_{2,b\bar{b}b\bar{b}}^{y_b y_b}. \tag{5.5}$$

The result of $\tilde{X}_{2,b\bar{b}}^{y_b y_b}$ is given by

$$\begin{aligned}
\tilde{X}_{2,b\bar{b}}^{y_b y_b}(w) = {} & -\frac{4\left(w^2+1\right)^2}{9\left(w^2-1\right)^2}\Big[100G(0,-1,-1,0,w)+28G(0,-1,0,-1,w) \\
& +14G(0,-1,0,1,w)+76G(0,-1,1,0,w)-128G(0,0,-1,-1,w)-64G(0,0,-1,1,w) \\
& -64G(0,0,1,-1,w)-32G(0,0,1,1,w)+76G(0,1,-1,0,w)-4G(0,1,0,-1,w) \\
& -2G(0,1,0,1,w)+35G(0,1,1,0,w)\Big]-\frac{32\left(w^4+10w^2+1\right)}{9(w-1)(w+1)^3}\Big[-6G(1,x_1,0,-1,w) \\
& +4G(1,x_1,0,0,w)-3G(1,x_1,0,1,w)+G(1,x_1,1,0,w)-6G(1,x_2,0,-1,w) \\
& +4G(1,x_2,0,0,w)-3G(1,x_2,0,1,w)+G(1,x_2,1,0,w)+6G(1,0,0,-1,w) \\
& +3G(1,0,0,1,w)-G(1,0,1,0,w)\Big]-\frac{32\left(2w^5+w^4+8w^3-2w^2+2w+1\right)}{9(w-1)^2(w+1)^3} \\
& \times\Big[6G(0,x_1,0,-1,w)-4G(0,x_1,0,0,w)+3G(0,x_1,0,1,w)-G(0,x_1,1,0,w) \\
& +6G(0,x_2,0,-1,w)-4G(0,x_2,0,0,w)+3G(0,x_2,0,1,w)-G(0,x_2,1,0,w)\Big]
\end{aligned}$$

JHEP03(2024)068

$$
\begin{aligned}
&+\frac{1}{9\left(w^2-1\right)^3}\Big[-8\left(w^4-w^3-16w^2-w+1\right)(w-1)^2G(1,0,0,0,w)\\
&-16w\left(6w^4+2w^3+31w^2-19w+4\right)(w-1)G(0,0,0,-1,w)\\
&+w\left(6w^5-30w^4-113w^3-194w^2-119w-30\right)G(0,0,0,0,w)\\
&-4w\left(12w^5+w^4+94w^3-46w^2+82w+1\right)G(0,0,0,1,w)\\
&+4\left(90w^6-96w^5+741w^4-1226w^3+523w^2-96w-128\right)G(0,-1,0,0,w)\\
&+32\left(6w^6+4w^5-35w^4+94w^3-33w^2+4w+8\right)G(0,0,-1,0,w)\\
&+2\left(94w^6+48w^5-431w^4+1118w^3-413w^2+48w+112\right)G(0,0,1,0,w)\\
&+2\left(29w^6-60w^5+959w^4-1628w^3+749w^2-60w-181\right)G(0,1,0,0,w)\Big]\\
&+\frac{\pi^4\left(231w^6-528w^5-1774w^4-5226w^3-1986w^2-528w+19\right)}{1620\left(w^2-1\right)^3}\\
&-\frac{4i\pi^3\left(w^4-w^3-16w^2-w+1\right)G(1,w)}{9(w-1)(w+1)^3}\\
&+\frac{256\pi^2\left(w^4+10w^2+1\right)\left(G(1,x_1,w)+G(1,x_2,w)\right)}{27(w-1)(w+1)^3}\\
&-\frac{256\pi^2\left(2w^5+w^4+8w^3-2w^2+2w+1\right)\left(G(0,x_1,w)+G(0,x_2,w)\right)}{27(w-1)^2(w+1)^3}\\
&+\frac{i\pi^3w\left(10w^5-18w^4-65w^3-26w^2-75w-18\right)G(0,w)}{27\left(w^2-1\right)^3}\\
&+\frac{2\pi^2}{27\left(w^2-1\right)^3}\Big[10\left(w^4-w^3-16w^2-w+1\right)(w-1)^2G(1,0,w)\\
&-w\left(8w^5-24w^4-89w^3-110w^2-97w-24\right)G(0,0,w)\\
&-\left(361w^6-384w^5+2956w^4-4698w^3+2084w^2-384w-511\right)G(0,-1,w)\\
&-3\left(19w^6-40w^5+642w^4-1154w^3+502w^2-40w-121\right)G(0,1,w)\Big]\\
&-\frac{32\left(w^4+10w^2+1\right)\zeta(3)G(1,w)}{3(w-1)(w+1)^3}\\
&-\frac{2\left(15w^6-18w^5-78w^4+386w^3-94w^2-18w-1\right)\zeta(3)G(0,w)}{9\left(w^2-1\right)^3}\\
&-\frac{32i\pi\left(w^4+10w^2+1\right)}{9(w-1)(w+1)^3}\Big[-4G(1,x_1,0,w)-G(1,x_1,1,w)-4G(1,x_2,0,w)\\
&-G(1,x_2,1,w)+G(1,0,1,w)\Big]-\frac{32i\pi\left(2w^5+w^4+8w^3-2w^2+2w+1\right)}{9(w-1)^2(w+1)^3}\\
&\times\Big[4G(0,x_1,0,w)+G(0,x_1,1,w)+4G(0,x_2,0,w)+G(0,x_2,1,w)\Big]\\
&+\frac{4i\pi\left(w^2+1\right)^2}{9\left(w^2-1\right)^2}\Big[100G(0,-1,-1,w)+76G(0,-1,1,w)+76G(0,1,-1,w)\\
&+35G(0,1,1,w)\Big]+\frac{\pi i}{9\left(w^2-1\right)^3}\\
&\times\Big[-w\left(6w^5-30w^4-113w^3-194w^2-119w-30\right)G(0,0,0,w)\\
&-4\left(90w^6-96w^5+741w^4-1226w^3+523w^2-96w-128\right)G(0,-1,0,w)
\end{aligned}
$$

JHEP03(2024)068

$$
\begin{aligned}
&-32\left(6w^6+4w^5-35w^4+94w^3-33w^2+4w+8\right)G(0,0,-1,w)\\
&-2\left(94w^6+48w^5-431w^4+1118w^3-413w^2+48w+112\right)G(0,0,1,w)\\
&-2\left(29w^6-60w^5+959w^4-1628w^3+749w^2-60w-181\right)G(0,1,0,w)\\
&+2\left(15w^6-18w^5-78w^4+386w^3-94w^2-18w-1\right)\zeta(3)\Big]\\
&+\frac{8i\pi\left(w^4-w^3-16w^2-w+1\right)G(1,0,0,w)}{9(w-1)(w+1)^3}\\
&+\frac{2(2n_l-55)(w^2+1)}{9\left(w^2-1\right)}\log\left(\frac{\mu^2}{m_H^2}\right)\Big[-2i\pi G(-1,w)-i\pi G(1,w)+2G(-1,0,w)\\
&-2G(0,-1,w)-G(0,1,w)+G(1,0,w)\Big]+\frac{4\left(w^2+1\right)}{9(w-1)(w+1)}\\
&\times\Big[(166-4n_l)G(-1,-1,0,w)+(94-4n_l)G(-1,0,-1,w)+(110-4n_l)G(-1,0,0,w)\\
&+(2n_l-63)G(-1,0,1,w)+(4n_l-34)G(-1,1,0,w)+(8n_l-260)G(0,-1,-1,w)\\
&+(4n_l-110)G(0,0,-1,w)+(2n_l-55)G(0,0,1,w)+(45-2n_l)G(0,1,1,w)\\
&+(4n_l-34)G(1,-1,0,w)+(110-4n_l)G(1,0,-1,w)+(55-2n_l)G(1,0,0,w)\\
&+(4n_l-110)G(1,1,0,w)-20G(0,-1,1,w)-20G(0,1,-1,w)\Big]\\
&+\frac{31w^4+86w^3-49w^2+86w+31}{9(w-1)(w+1)^3}\Big[6G(x_1,0,-1,w)-4G(x_1,0,0,w)+3G(x_1,0,1,w)\\
&-G(x_1,1,0,w)+6G(x_2,0,-1,w)-4G(x_2,0,0,w)+3G(x_2,0,1,w)-G(x_2,1,0,w)\Big]\\
&+\frac{4n_l\left(w^5+w^4-w-1\right)-118w^5+480w^4-459w^3+331w^2+628w+162}{9(w-1)^2(w+1)^3}\\
&\times\Big[2G(0,0,-1,w)-G(0,0,1,w)\Big]+\frac{1}{18(w-1)^2(w+1)^3}\\
&\times\Big[2(w+1)\left(20n_l\left(w^4-1\right)-383w^4+1586w^3-2786w^2+1586w+1149\right)\\
&\times G(-1,0,0,w)-32(w+1)\left(n_l\left(w^4-1\right)+3w^4+32w^3-123w^2+32w+40\right)\\
&\times G(0,-1,0,w)+8\left(-4n_l(w+1)^2\left(w^2+1\right)+63w^4+120w^3+130w^2+120w+63\right)\\
&\times(w-1)G(1,0,-1,w)\\
&-\left(-24n_l(w+1)^2\left(w^2+1\right)+183w^4-1220w^3+1662w^2+4444w+1459\right)\\
&\times(w-1)G(1,0,0,w)\\
&+2\left(-8n_l(w+1)^2\left(w^2+1\right)+126w^4+240w^3+260w^2+240w+126\right)\\
&\times(w-1)G(1,0,1,w)\\
&-8\left(4n_l\left(w^5+w^4-w-1\right)-118w^5+480w^4-459w^3+331w^2+628w+162\right)\\
&\times G(0,0,-1,w)\\
&-2\left(12n_l\left(w^5+w^4-w-1\right)+89w^5+319w^4-1018w^3-1278w^2+760w+488\right)\\
&\times G(0,1,0,w)+2(w-1)\left(139w^4+284w^3+274w^2+284w+139\right)G(1,1,0,w)\\
&+\left(-11w^5-43w^4-203w^3-157w^2+26w+4\right)G(0,0,0,w)\Big]\\
&+\frac{8i\pi\left(w^2+1\right)}{9\left(w^2-1\right)}\Big[(2n_l-83)G(-1,-1,w)-(2n_l-17)G(-1,1,w)
\end{aligned}
$$

JHEP03(2024)068

$$
\begin{aligned}
&-(2n_l-17)G(1,-1,w)\Big]+\frac{i\pi\left(31w^4+86w^3-49w^2+86w+31\right)}{9(w-1)(w+1)^3}\Big[4G(x_1,0,w)\\
&+G(x_1,1,w)+4G(x_2,0,w)+G(x_2,1,w)\Big]\\
&+\frac{8\pi^2\left(31w^4+86w^3-49w^2+86w+31\right)}{27(w-1)(w+1)^3}\Big[G(x_1,w)+G(x_2,w)\Big]\\
&-\frac{i\pi^3\left(22w^5+66w^4+205w^3+131w^2-37w-3\right)}{54(w-1)^2(w+1)^3}+\frac{2\left(w^2-4w+1\right)\pi^2\log(2)}{9(w+1)^2}\\
&-\frac{\left(6n_l\left(w^5+w^4-w-1\right)-159w^5+317w^4+44w^3+844w^2+510w+172\right)\zeta(3)}{9(w-1)^2(w+1)^3}\\
&+\frac{\pi^2}{54(w-1)^2(w+1)^3}\Big[-(w+1)(32n_l\left(w^4-1\right)-217w^4+6354w^3-10770w^2\\
&+6354w+3271)G(-1,w)+\left(33w^5+109w^4+408w^3+288w^2-63w-7\right)G(0,w)\\
&-\left(24n_l\left(w^5+w^4-w-1\right)+311w^5+3473w^4-6172w^3-5972w^2+5317w+2275\right)\\
&\times G(1,w)\Big]\\
&+\frac{i\pi}{18(w-1)^2(w+1)^3}\Big[-2(w+1)\\
&\times(4n_l\left(w^4-1\right)+57w^4+1586w^3-2786w^2+1586w+709)G(-1,0,w)\\
&+32(w+1)\left(n_l\left(w^4-1\right)+3w^4+32w^3-123w^2+32w+40\right)G(0,-1,w)\\
&+\left(11w^5+43w^4+203w^3+157w^2-26w-4\right)G(0,0,w)\\
&+2\left(12n_l\left(w^5+w^4-w-1\right)+89w^5+319w^4-1018w^3-1278w^2+760w+488\right)\\
&\times G(0,1,w)\\
&-\left(8n_l\left(w^5+w^4-w-1\right)+257w^5+1843w^4-2882w^3-2782w^2+2545w+1019\right)\\
&\times G(1,0,w)\\
&+\left(-32n_l\left(w^5+w^4-w-1\right)+602w^5+590w^4+20w^3-20w^2-590w-602\right)\\
&\times G(1,1,w)\Big]-\frac{2n_l-55}{12}\log^2\left(\frac{\mu^2}{m_H^2}\right)+\frac{2n_l-55}{9}\log\left(\frac{\mu^2}{m_H^2}\right)\Big(4G(-1,w)-G(1,w)\Big)\\
&+\frac{i(2n_l-55)w\left(3w^3-w^2-8w-7\right)}{9(w-1)(w+1)^3}\log\left(\frac{\mu^2}{m_H^2}\right)\Big(\pi+iG(0,w)\Big)\\
&-\frac{2}{9}\Big[16n_lG(-1,-1,w)+2n_lG(1,1,w)-392G(-1,-1,w)+24G(-1,1,w)\\
&+24G(1,-1,w)-43G(1,1,w)\Big]-\frac{2w}{3\left(w^2-1\right)}\Big[-2G(0,0,w)+2i\pi G(0,w)\Big]\\
&+\frac{8\left(n_l\left(11w^2-12w-1\right)-321w^2+471w-45\right)}{27\left(w^2-1\right)}G(-1,0,w)\\
&+\frac{1}{27(w-1)\left(w^3+1\right)^3}\Big[\Big(4n_l\left(w^2-w+1\right)^3\left(11w^4+4w^3-20w^2-44w-13\right)\\
&-6(41w^{10}-626w^9+957w^8-2300w^7+2753w^6-4566w^5+3857w^4-3404w^3\\
&+957w^2-258w-327)\Big)G(0,-1,w)-\Big(2n_l(w+1)^2\left(w^2-w+1\right)^3\left(7w^2-24w+13\right)\\
&-3(289w^{10}-450w^9+449w^8+1224w^7-2925w^6+4408w^5-3039w^4+1338w^3
\end{aligned}
$$

JHEP03(2024)068

$$
\begin{aligned}
&+449w^2-488w+327)\Big)G(0,1,w)-\Big(4n_l\left(w^2-w+1\right)^3\left(w^4-w^3-28w^2+5w+4\right)\\
&-3(57w^{10}-245w^9-516w^8+2349w^7-5403w^6+6486w^5-5517w^4+2463w^3\\
&-516w^2-283w+95)\Big)G(1,0,w)\Big]+\frac{1}{54(w-1)^2(w+1)^4\left(w^2-w+1\right)^3}\\
&\times\Big[-2(6n_l+527)w^{12}+(336n_l-12575)w^{11}+(32735-936n_l)w^{10}\\
&+24(57n_l-2240)w^9+(31063-756n_l)w^8+(12305-684n_l)w^7\\
&+4(423n_l-12790)w^6+(44291-1440n_l)w^5+(360n_l-15539)w^4\\
&+24(13n_l-174)w^3+(6221-348n_l)w^2+(108n_l-1061)w+38\Big]G(0,0,w)\\
&+\frac{\pi^2}{54(w-1)^2(w+1)^4\left(w^2-w+1\right)^3}\Big[n_l\left(w^2-w+1\right)^3(23w^6-162w^5-71w^4+164w^3\\
&+45w^2-2w+3)+281w^{12}+9318w^{11}-21794w^{10}+34216w^9-18084w^8-7596w^7\\
&+31012w^6-26382w^5+11106w^4-160w^3-1732w^2-84w-117\Big]\\
&-\frac{8i\pi\left(n_l\left(11w^2-12w-1\right)-321w^2+471w-45\right)}{27\left(w^2-1\right)}G(-1,w)\\
&+\frac{i\pi}{54(w-1)^2(w+1)^4\left(w^2-w+1\right)^3}\Big[2(6n_l+527)w^{12}+(12575-336n_l)w^{11}\\
&+(936n_l-32735)w^{10}-24(57n_l-2240)w^9+(756n_l-31063)w^8+(684n_l-12305)w^7\\
&+(51160-1692n_l)w^6+(1440n_l-44291)w^5+(15539-360n_l)w^4+(4176-312n_l)w^3\\
&+(348n_l-6221)w^2+(1061-108n_l)w-38\Big]G(0,w)\\
&+\frac{i\pi}{27(w-1)\left(w^3+1\right)^3}\Big[4n_l\left(w^2-w+1\right)^3\left(w^4-w^3-28w^2+5w+4\right)\\
&-3(57w^{10}-245w^9-516w^8+2349w^7-5403w^6+6486w^5-5517w^4\\
&+2463w^3-516w^2-283w+95)\Big]G(1,w)\\
&-\frac{\left(n_l\left(22w^2+50w+22\right)-614w^2-1393w-614\right)}{18(w+1)^2}\log\left(\frac{\mu^2}{m_H^2}\right)\\
&+\frac{1}{27\left(w^3+1\right)^2}\Big[2n_l\left(w^2-w+1\right)^2\left(59w^2+136w+59\right)\\
&-3\left(993w^6+736w^5-1424w^4+4130w^3-1424w^2+736w+993\right)\Big]G(-1,w)\\
&+\frac{1}{324(w-1)(w+1)^3\left(w^2-w+1\right)^2}\Big[(31015-1332n_l)w^8+6(170n_l-2271)w^7\\
&+(1440n_l-81929)w^6+(192374-5112n_l)w^5+6(916n_l-34731)w^4\\
&-22(72n_l-2963)w^3+(14605-756n_l)w^2+6(242n_l-7905)w-503\Big]G(0,w)\\
&+\frac{1}{54(w+1)^2\left(w^2-w+1\right)^2}\Big[-14n_l\left(w^3+1\right)^2\\
&+3\left(235w^6-406w^5+764w^4-684w^3+764w^2-406w+235\right)\Big]G(1,w)\\
&+\frac{i\pi}{324(w-1)(w+1)^3\left(w^2-w+1\right)^2}\Big[(1332n_l-31015)w^8
\end{aligned}
$$

JHEP03(2024)068

$$
\begin{aligned}
&-6(170n_l-2271)w^7+(81929-1440n_l)w^6+2(2556n_l-96187)w^5\\
&-6(916n_l-34731)w^4+22(72n_l-2963)w^3+(756n_l-14605)w^2\\
&-6(242n_l-7905)w+503\Big]\\
&+\frac{-18n_l\left(195w^2+374w+195\right)+98681w^2+204550w+98681}{1296(w+1)^2}
\end{aligned}
\tag{5.6}
$$

with

$$
x_1 = l_4^{\mathrm{NP1}} = \frac{1+\sqrt{3}i}{2}, \quad x_2 = l_5^{\mathrm{NP1}} = \frac{1-\sqrt{3}i}{2}\,. \tag{5.7}
$$

Here the color factors $C_F = 4/3$, $C_A = 3$, and $T_R = 1/2$ have been substituted by their numeric values, and n_l denotes the number of light quark flavors.

The result of $X^{y_b y_b}_{2,b\bar{b}b\bar{b}}$ is given by

$$
\begin{aligned}
X^{y_b y_b}_{2,b\bar{b}b\bar{b}} = \frac{8}{3\pi\beta(z-4)z}\Bigg[&\frac{F_1^{4b}(z)\left(-207z^2-3964z-11264\right)}{55296}\\
&+\frac{F_2^{4b}(z)\left(479z^2+18860z-34528\right)}{82944}-\frac{F_3^{4b}(z)(z-4)(2491z+3012)}{20736}\\
&+\frac{F_4^{4b}(z)\left(-55z^3+254z^2+2328z-9216\right)}{3456\beta z}+\frac{F_5^{4b}(z)\left(-5z^2+24z-32\right)}{2304}\\
&+\frac{\beta F_6^{4b}(z)\left(-5z^2+10z+384\right)}{1152}+\frac{F_7^{4b}(z)\left(5z^2-42z+90\right)}{576}\\
&+\frac{\beta F_8^{4b}(z)\left(17z^2-58z-384\right)}{1152}+\frac{F_9^{4b}(z)\left(z^3-6z^2+z-192\right)}{288z}\\
&+\frac{F_{10}^{4b}(z)\left(z^2-3z-16\right)}{144}-\frac{\beta F_{12}^{4b}(z)(z-2)^2}{144}\Bigg]\,,
\end{aligned}
\tag{5.8}
$$

where $\beta = \sqrt{1-4/z}$ measures the velocity of the bottom quark. The master integrals $F_1^{4b}(z)$, $F_2^{4b}(z)$ and $F_3^{4b}(z)$ are linear combinations of complete elliptic integrals or their derivatives, and the other $F_i^{4b}(z)$ can be expressed by one-fold integrals of MPLs and complete elliptic integrals. The complete expressions of all $F_i^{4b}(z)$ can be found in [53]. We also provide these expressions in electronic form in the supplementary material attached to this paper.

5.2 Asymptotic expansion

It is interesting to explore the decay width in various limits. First, we study the small bottom quark mass limit, i.e., $m_b^2 \to 0$ or $z \to \infty$, while the Yukawa coupling is still kept finite. The NLO coefficient becomes

$$
X_1^{y_b y_b}|_{z\to\infty} = 2\log\left(\frac{\mu^2}{m_H^2}\right)+\frac{17}{3}+\frac{12\log(z)+10}{z}+\mathcal{O}(z^{-2})\,. \tag{5.9}
$$

Therefore, it is safe to take the continuous limit of $m_b = 0$. This property is in accordance with the Kinoshita-Lee-Nauenberg theorem [78, 79]. Here the parameter m_b can be considered as a regulator for infrared divergences. Though the virtual and real corrections contain $\log(z)$ terms induced by collinear gluons separately, the dependence on the regulator cancels in

JHEP03(2024)068

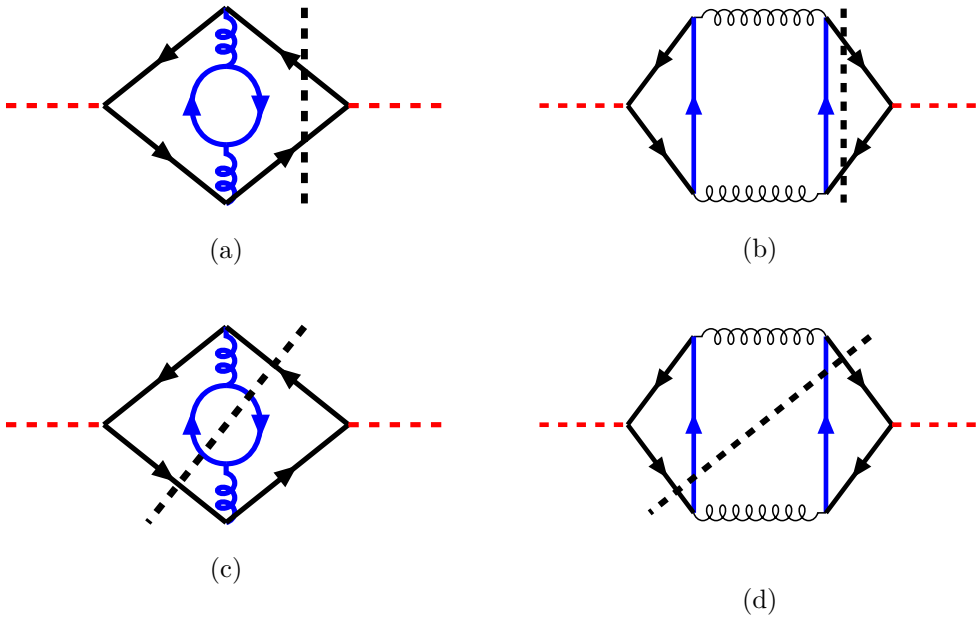

Figure 8. Typical cut diagrams that have large logarithms $\log^i(z)$. The thick black and red lines stand for the massive bottom quark and the Higgs boson, respectively. And the blue line represents the particle that is soft.

the decay width at $\mathcal{O}(z^0)$. Notice that there are large $\log(z)$ terms at $\mathcal{O}(z^0)$ if the Yukawa coupling is renormalized in the on-shell scheme.

In the limit of $z \to \infty$, the NNLO coefficient $\tilde{X}^{y_b y_b}_{2,b\bar{b}}$ reads

$$\begin{aligned}
\tilde{X}^{y_b y_b}_{2,b\bar{b}}|_{z\to\infty} = & -\frac{\log^3(z)}{27}+\frac{19\log^2(z)}{54}+\frac{2\zeta(3)\log(z)}{9}+\frac{\pi^2\log(z)}{54} \\
& -\frac{503\log(z)}{324}+\frac{55-2n_l}{12}\log^2\left(\frac{\mu^2}{m_H^2}\right)+\frac{307-11n_l}{9}\log\left(\frac{\mu^2}{m_H^2}\right)+\frac{2(3n_l-86)\zeta(3)}{9} \\
& +\frac{(3n_l-98)\pi^2}{54}-\frac{19\pi^4}{1620}+\frac{2\log(2)\pi^2}{9}+\frac{98681-3510n_l}{1296} \\
& +\frac{1}{z}\Bigg(-\frac{5\log^4(z)}{36}+\frac{\log^3(z)}{18}+\frac{\pi^2\log^2(z)}{18}-\frac{(2n_l-9)\log^2(z)}{2}-\frac{7\pi^2\log(z)}{54} \\
& -(2n_l-55)\log(z)\log\left(\frac{\mu^2}{m_H^2}\right)-\frac{(117n_l-3625)\log(z)}{27}-4\zeta(3)\log(z) \\
& -\frac{71\pi^4}{540}+\frac{4\pi^2\log(2)}{3}-\frac{5(2n_l-55)}{6}\log\left(\frac{\mu^2}{m_H^2}\right)+\frac{430\zeta(3)}{9}-\frac{(9n_l-175)\pi^2}{9} \\
& +\frac{36829-1548n_l}{324}\Bigg)+\mathcal{O}(z^{-2}),
\end{aligned} \tag{5.10}$$

where we have shown explicitly the first two orders in the expansion. It is obvious that there are large logarithms $\log^i(z)$ with $i \leq 3$ at $\mathcal{O}(z^0)$. This means that the contribution of this part with two bottom quarks in the final state to the decay width is divergent in the massless limit.

These logarithms $\log^i(z)$ arise from the diagrams with a bottom quark loop, e.g., see the diagram (a) in figure 8. These diagrams contain up to $1/\epsilon^3$ infrared divergences in the massless case, which are converted to $\log^i(z), i \leq 3$, in the massive bottom quark case.

JHEP03(2024)068

It is interesting to note the $\log^4(z)$ term at order $\mathcal{O}(z^{-1})$. It appears due to a different reason. It comes from the contribution of diagrams with two soft quarks exchanged between two separate directions, e.g., see the diagram (b) in figure 8.

The other NNLO coefficient $X^{y_b y_b}_{2,b\bar{b}b\bar{b}}$ consists of master integrals that depend on elliptic integrals. We present the asymptotic expansion results of the master integrals in appendix B. Based on these results, we obtain the asymptotic expansion,

$$\begin{aligned}
X^{y_b y_b}_{2,b\bar{b}b\bar{b}}|_{z\to\infty} = & \frac{\log^3(z)}{27} - \frac{19\log^2(z)}{54} - \frac{2\zeta(3)\log(z)}{9} - \frac{\pi^2\log(z)}{54} + \frac{503\log(z)}{324} \\
& + \frac{19\pi^4}{1620} - \frac{2\log(2)\pi^2}{9} + \frac{5\zeta(3)}{18} + \frac{31\pi^2}{108} - \frac{2491}{648} \\
& + \frac{1}{z}\left(\frac{\log^4(z)}{18} - \frac{\log^3(z)}{18} - \frac{2\pi^2\log^2(z)}{9} + \frac{7\pi^2\log(z)}{54} - \frac{\log^2(z)}{3} + 4\zeta(3)\log(z)\right. \\
& \left. + \frac{211\log(z)}{54} + \frac{13\pi^4}{270} - \frac{7\zeta(3)}{9} + \frac{19\pi^2}{18} - \frac{8353}{324}\right) + \mathcal{O}(z^{-2})\,. \qquad (5.11)
\end{aligned}$$

It can be seen that $X^{y_b y_b}_{2,b\bar{b}b\bar{b}}$ also have large logarithms when $z \to \infty$. The logarithms $\log^i(z), i \leq 3$, at $\mathcal{O}(z^0)$ are obtained by calculating the diagrams with a soft gluon splitting to a soft bottom quark pair; see the diagram (c) in figure 8. In contrast, the logarithm $\log^4(z)$ at $\mathcal{O}(z^{-1})$ arises from the diagrams with a collinear quark splitting to a soft quark and a collinear gluon which splits into a collinear quark and a soft quark further, such as the diagram (d) in figure 8.

Combining the results in eq. (5.10) and eq. (5.11), we obtain

$$\begin{aligned}
X^{y_b y_b}_2|_{z\to\infty\,(m_b^2\to 0)} = & \frac{55-2n_l}{12}\log^2\left(\frac{\mu^2}{m_H^2}\right) + \frac{307-11n_l}{9}\log\left(\frac{\mu^2}{m_H^2}\right) + \frac{(4n_l-113)\zeta(3)}{6} \\
& + \frac{(2n_l-55)\pi^2}{36} + \frac{10411-390n_l}{144} + \frac{1}{z}\left(-\frac{\log^4(z)}{12} - \frac{\pi^2\log^2(z)}{6} - \frac{(6n_l-25)\log^2(z)}{6}\right. \\
& - \frac{(26n_l-829)\log(z)}{6} - (2n_l-55)\log(z)\log\left(\frac{\mu^2}{m_H^2}\right) - \frac{5(2n_l-55)}{6}\log\left(\frac{\mu^2}{m_H^2}\right) - \frac{\pi^4}{12} \\
& \left. + \frac{4\pi^2\log(2)}{3} + 47\zeta(3) - \frac{(2n_l-41)\pi^2}{2} + \frac{791-43n_l}{9}\right) + \mathcal{O}(z^{-2})\,. \qquad (5.12)
\end{aligned}$$

The large logarithms of z at $\mathcal{O}(z^0)$ cancel, as expected. However, the large logarithms at $\mathcal{O}(z^{-1})$ survive. The structure of this kind of large logarithm at higher orders in α_s is not well understood. It is promising to explore this issue using the effective field theory, which we leave for future study. After adopting the same renormalization scheme, the above expression is consistent with the result in ref. [45], which was obtained with the method of large momentum expansion.

It is also interesting to investigate the threshold limit of bottom quark pair production, where the coefficients $X_1^{y_b y_b}$ and $X_2^{y_b y_b}$ are expanded in the limit of $m_H \to 2m_b$ or $\beta \to 0$. Expanding the full analytic results, we have

$$X_1^{y_b y_b}|_{\beta\to 0} = \frac{2\pi^2}{3\beta} + 4\log(2) + \frac{4}{3} + 2\log\left(\frac{\mu^2}{m_H^2}\right) + \frac{2\pi^2\beta}{3} + \mathcal{O}\left(\beta^2\right), \qquad (5.13)$$

JHEP03(2024)068

and

$$
\begin{aligned}
X_2^{y_b y_b}|_{\beta\to 0} = & \frac{1}{\beta^2}\left(\frac{4\pi^4}{27}+\frac{4\pi^2}{9}\right)+\frac{1}{\beta}\Bigg(\frac{(339-22n_l)\,\pi^2}{54}+\frac{22\log(2)\pi^2}{9} \\
& +\frac{(2n_l-33)\,\pi^2\log(\beta)}{9}+\frac{(55-2n_l)\,\pi^2}{18}\log\left(\frac{\mu^2}{m_H^2}\right)\Bigg) \\
& +\Bigg(\frac{8\pi^4}{27}+\frac{(139-3n_l)\pi^2}{27}-\frac{100\log(\beta)\pi^2}{27}-\frac{130\log(2)\pi^2}{27}-\frac{181\zeta(3)}{9} \\
& +\frac{(55-2n_l)}{12}\log^2\left(\frac{4\mu^2}{m_H^2}\right)+\frac{(57-2n_l)}{4}\log\left(\frac{\mu^2}{m_H^2}\right)+\frac{(163-6n_l)\log(2)}{6} \\
& +\frac{(13993-506n_l)}{432}\Bigg)+\beta\Bigg(\frac{(6n_l-303)\,\pi^2}{162}+\frac{(6n_l+101)\,\pi^2\log(\beta)}{27} \\
& +\frac{(55-2n_l)\,\pi^2}{18}\log\left(\frac{\mu^2}{m_H^2}\right)+\frac{418\log(2)\pi^2}{27}\Bigg)+\mathcal{O}\left(\beta^2\right). \qquad (5.14)
\end{aligned}
$$

In obtaining the above compact expressions, we have extensively used the relations among MPLs in [80, 81]. We check that the coefficients of $1/\beta^2$ and $1/\beta$ are consistent with the results of two-loop form factors [82], which means that the power divergences at $\beta \to 0$ in $X_1^{y_b y_b}$ and $X_2^{y_b y_b}$ come only from virtual corrections. Actually, these velocity-enhanced terms are due to the Coulomb potential interaction between the bottom quark pair. They violate the convergence of perturbative expansion[2] and need to be resummed to all orders in α_s by calculating the corresponding Coulomb Green function [83]. Notice that the coefficients of π^2/β in $X_1^{y_b y_b}$ and π^4/β^2 in $X_2^{y_b y_b}$ agree with those in $e^+e^- \to t\bar{t}$ production near threshold [84], but the π^2/β^2 term is different. They can be reproduced by calculating the imaginary part of the first a few terms of the zero-distance Coulomb Green function,

$$
\begin{aligned}
\mathrm{Im}\ G_0(E) = & -\mathrm{Im}\int\frac{d^{D-1}\mathbf{p}_1}{(2\pi)^{D-1}}\frac{\mathbf{p}_1\cdot\mathbf{p}_1}{E-\frac{\mathbf{p}_1^2}{m_b}}+\mathrm{Im}\int\frac{d^{D-1}\mathbf{p}_1}{(2\pi)^{D-1}}\frac{d^{D-1}\mathbf{p}_2}{(2\pi)^{D-1}}\frac{\mathbf{p}_1\cdot\mathbf{p}_2}{E-\frac{\mathbf{p}_1^2}{m_b}}\frac{g_s^2C_F}{(\mathbf{p}_1-\mathbf{p}_2)^2}\frac{1}{E-\frac{\mathbf{p}_2^2}{m_b}} \\
& -\mathrm{Im}\int\frac{d^3\mathbf{p}_1}{(2\pi)^3}\frac{d^3\mathbf{p}_2}{(2\pi)^3}\frac{d^3\mathbf{p}_3}{(2\pi)^3}\frac{\mathbf{p}_1\cdot\mathbf{p}_3}{E-\frac{\mathbf{p}_1^2}{m_b}}\frac{g_s^2C_F}{(\mathbf{p}_1-\mathbf{p}_2)^2}\frac{1}{E-\frac{\mathbf{p}_2^2}{m_b}}\frac{g_s^2C_F}{(\mathbf{p}_2-\mathbf{p}_3)^2}\frac{1}{E-\frac{\mathbf{p}_3^2}{m_b}}+\mathcal{O}(g_s^6) \\
= & \frac{\beta^3m_b^4}{4\pi}\left(1+\frac{\alpha_sC_F}{\pi}\frac{\pi^2}{2\beta}+\left(\frac{\alpha_sC_F}{\pi}\right)^2\left(\frac{\pi^4}{12\beta^2}+\frac{\pi^2}{4\beta^2}\right)+\mathcal{O}(\alpha_s^3)\right) \qquad (5.15)
\end{aligned}
$$

with $E = m_b\beta^2 + \mathcal{O}(\beta^4)$. The $\mathbf{p}_i \cdot \mathbf{p}_j$ numerator arises because the scalar current $\bar{\psi}_b\psi_b$ with ψ_b the bottom quark field matches to the current $j^{(s)} = \psi^\dagger\boldsymbol{\sigma}\cdot i\mathbf{D}\chi/m_b$ in non-relativistic QCD where ψ and χ represent the two-component quark and anti-quark field, respectively. The all-order result of this Coulomb Green function can be found in refs. [85, 86].

[2]The perturbative expansion has a structure $\beta^3\sum_{n=0}c_n\alpha_s^n/\beta^n$ in the limit of $\beta \to 0$. It becomes divergent for $n > 3$.

JHEP03(2024)068

μ	$\frac{1}{2}m_H$	m_H	$2m_H$
$X_1^{y_b y_b}$(our res.)	+3.037513882	+5.810102605	+8.582691327
$X_1^{y_b y_b}$(Ref. [45])	+3.037513882	+5.810102605	+8.582691327
$\tilde{X}_{2,b\bar{b}}^{y_b y_b}$(our res.)	−4.537634223	+29.22467780	+78.04118427
$X_2^{y_b y_b}$(our res.)	−3.181291730	+30.58102030	+79.39752676
$X_2^{y_b y_b}$(Ref. [45])	−3.181291730	+30.58102030	+79.39752676

Table 1. The numerical results for the $X_1^{y_b y_b}$, $\tilde{X}_2^{y_b y_b}$ and $X_2^{y_b y_b}$ coefficients.

6 Numerical results

We adopt the SM input parameters [3]

$$\begin{aligned} \overline{m_b}\,(\overline{m_b}) &= 4.18 \text{ GeV}, & m_b &= 5.07 \text{ GeV}, \\ m_H &= 125.09 \text{ GeV}, & m_Z &= 91.1876 \text{ GeV}, \\ \alpha_s(m_Z) &= 0.1181, & G_F &= 1.166378 \times 10^{-5} \text{ GeV}^{-2}, \end{aligned} \tag{6.1}$$

where the on-shell value of m_b is obtained from $\overline{m_b}(\overline{m_b})$ in the $\overline{\text{MS}}$ scheme with the four-loop mass relation. The package `RunDec` [87, 88] was used to obtain $\overline{m_b}$ at other scales, e.g., $\overline{m_b}\,(m_H/2) = 2.956\,\text{GeV}$, $\overline{m_b}\,(m_H) = 2.784\,\text{GeV}$ and $\overline{m_b}\,(2m_H) = 2.639\,\text{GeV}$.

The numerical results of $X_1^{y_b y_b}$ and $X_2^{y_b y_b}$ at the scale $\mu = m_H/2, m_H, 2m_H$ are shown in table 1. We also provide the results from the asymptotic expansion expression[3] in [45], which have been calculated up to $\mathcal{O}(m_b^8/m_H^8)$. Perfect agreements were found for both the NLO $X_1^{y_b y_b}$ and NNLO $X_2^{y_b y_b}$ coefficients. The reason is that the expansion parameter $m_b^2/m_H^2 \approx 5 \times 10^{-4}$ is rather small. Actually, the difference between the expansion up to $\mathcal{O}(m_b^2/m_H^2)$ and the exact result at the scale $\mu = m_H$ is already 0.002% for $X_2^{y_b y_b}$.

Combining the corrections up to $\mathcal{O}(y_b^2\alpha_s^2)$, we obtain $\Gamma_{Hb\bar{b}}^{y_b y_b}$ up to NNLO. We show the results of $\Gamma_{Hb\bar{b}}^{y_b y_b}$ with both the $\overline{\text{MS}}$ and the on-shell Yukawa coupling in figure 9. The renormalization scale μ varies from $\mu = 1/2m_H$ to $\mu = 2m_H$. In the $\overline{\text{MS}}$ scheme, the LO decay width exhibits a strong scale dependence, and it changes by about 23% for μ in the range $[1/2m_H, 2m_H]$. The scale uncertainty is significantly reduced down to about 9% and 3% at NLO and NNLO, respectively. The NLO correction increases the LO decay width by a factor of 12% − 28% depending on the scale. The NNLO correction is very small at $\mu = m_H/2$ but can be as large as 7% at $\mu = 2m_H$.

The LO decay width in the on-shell scheme does not depend on the scale. The scale uncertainty is about 11% and 13% at NLO and NNLO, respectively. At a typical scale $\mu = m_H$, the NLO correction decreases the LO result by 35%, and the NNLO correction reduces the decay width further by 23%. This behavior indicates that the perturbative expansion with the on-shell Yukawa coupling converges slower than that with the $\overline{\text{MS}}$ Yukawa coupling.

[3]The result was calculated with the Yukawa coupling in the on-shell scheme. We converted it to the result in the $\overline{\text{MS}}$ scheme with the mass relation in [74–77].

JHEP03(2024)068

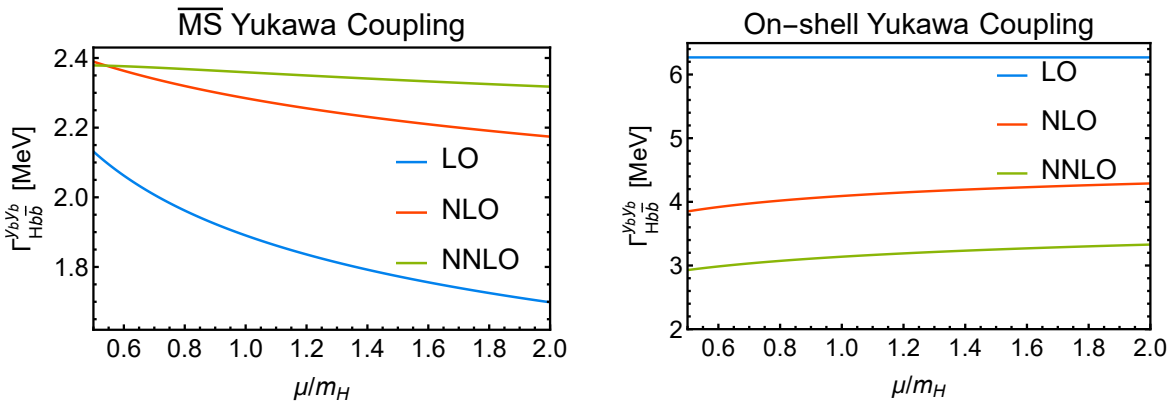

Figure 9. $\Gamma^{y_b y_b}_{Hb\bar{b}}$ at different values of μ. The left (right) plot corresponds to the result with $\overline{\text{MS}}$ (on-shell) Yukawa couplings. The LO, NLO and NNLO predictions are represented by the blue, red, and green lines, respectively.

width	LO	NLO	NNLO(y_b^2)	NNLO($y_b y_t$)	N^{3}LO(y_b^2)	N^{4}LO(y_b^2)
$\overline{\text{MS}}$	$1.891^{+0.241}_{-0.192}$	$2.285^{+0.105}_{-0.110}$	$2.359^{+0.020}_{-0.041}$	$2.376^{+0.026}_{-0.046}$	$2.379^{+0.005}_{-0.015}$	$2.377^{+0.006}_{-0.006}$
on-shell	6.269	$4.092^{+0.197}_{-0.242}$	$3.138^{+0.191}_{-0.210}$	$3.193^{+0.181}_{-0.197}$	$2.804^{+0.112}_{-0.096}$	$2.649^{+0.065}_{-0.049}$

Table 2. Decay width of $\Gamma_{Hb\bar{b}}$ (in MeV) at different orders in QCD. The second row shows the results with the $\overline{\text{MS}}$ Yukawa coupling while the third row shows the results with the on-shell Yukawa coupling. The renormalization scale uncertainties around $\mu = m_H$ are also shown. The NNLO($y_b y_t$), N^{3}LO(y_b^2), and N^{4}LO(y_b^2) results were calculated using the formulas in refs. [26–28, 89].

Comparing the two schemes, one can find that the results in the on-shell scheme are larger than those in the $\overline{\text{MS}}$ scheme. The main reason is that the pole mass $m_b = 5.07\,\text{GeV}$ is much larger than the $\overline{\text{MS}}$ mass $\overline{m_b}(\mu = m_H) = 2.78\,\text{GeV}$. However, higher-order corrections reduce the difference between the two schemes prominently, signifying its importance in providing reliable theoretical predictions.

Given that the scheme dependence constitutes the dominant theoretical uncertainty now, it is necessary to see how it changes as higher orders are taken into account. We adopt the $\mathcal{O}(y_b y_t \alpha_s^2)$ correction in the expansion of m_b^2/m_H^2 and m_H^2/m_t^2 [89], and the $\mathcal{O}(y_b^2 \alpha_s^3)$ and $\mathcal{O}(y_b^2 \alpha_s^4)$ corrections in the massless bottom quark limit [26–28]. One can see that the central value for the decay width in the $\overline{\text{MS}}$ scheme becomes stable quickly and that the scale uncertainty is decreased to $\pm 0.3\%$ after the N^{4}LO correction is included. In contrast, the N^{3}LO correction in the on-shell scheme decreases the NNLO result by 12%, and the N^{4}LO correction reduces the decay width by 5.5% further. The scale uncertainty is still 2% at N^{4}LO. All these features manifest that the results in the on-shell scheme have a bad convergence behavior. It is more appropriate to take the $\overline{\text{MS}}$ scheme in this high-energy process.

Our analytic results can also be applied to the decay of heavy scalar particles. If there is a new scalar Higgs with a mass of more than 350 GeV, it can decay to a top quark pair. In this process, we have five massless light quarks and one massive top quark. The decay width $\Gamma_{Ht\bar{t}}$ of $H \to t\bar{t}$ with $m_t = 172.69\,\text{GeV}$ and different values of m_H are shown in figure 10. The NNLO correction increases the NLO result by $3\% - 17\%$ at $\mu = m_H$ as m_H varies from 400 GeV to 1 TeV, and the scale uncertainty reduces from $7\% - 13\%$ at NLO to $2\% - 8\%$ at NNLO.

JHEP03(2024)068

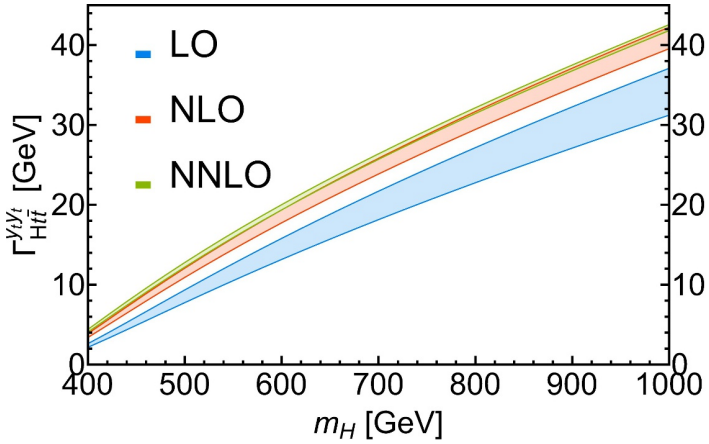

Figure 10. $\Gamma_{Ht\bar{t}}$ at different values of m_H. The LO, NLO and NNLO predictions with QCD scale uncertainties are represented by the blue, red, and green bands, respectively.

7 Conclusion

We present an exact result of the decay width of $H \to b\bar{b}$ at $\mathcal{O}(y_b^2\alpha_s^2)$ with full dependence on m_b. The result is obtained by calculating the imaginary part of the three-loop self-energy diagrams of the Higgs boson which have two bottom quarks in the propagators. The master integrals are calculated using the method of differential equations combined with the choice of a canonical basis. The result has been expressed in terms of multiple polylogarithms. The contribution from the four bottom quarks is also calculated analytically and written as a linear combination of complete elliptic integrals and one-fold integrals of them. The expansion of our analytical result in the small quark mass m_b limit agrees with the previous computation in a different method. The threshold expansion shows velocity enhancement which is caused by Coulomb potential interactions. Though the structure is not the same as that in the top quark pair production near the threshold, we checked that the power divergent terms can be reproduced by calculating a new Coulomb Green function. We discuss the numerical results in both the $\overline{\text{MS}}$ and on-shell renormalization schemes for the Yukawa coupling. Higher-order QCD corrections reduce both the scale uncertainty and the difference between the two schemes. The analytical results presented in this paper enable accurate and fast evaluation of the decay width of heavy scalar particles.

Acknowledgments

We would like to thank Long Chen, Long-Bin Chen, Hai Tao Li, Xing Wang and Yu-Ming Wang for helpful discussions. We thank Xin Guan for the communication on the implementation of packages `AMFlow` and `CalcLoop`. This work was supported in part by the National Science Foundation of China (grant Nos. 12005117, 12321005, 12375076) and the Taishan Scholar Foundation of Shandong province (tsqn201909011). The topology diagrams in this paper were drawn using the TikZ-Feynman package [90].

JHEP03(2024)068

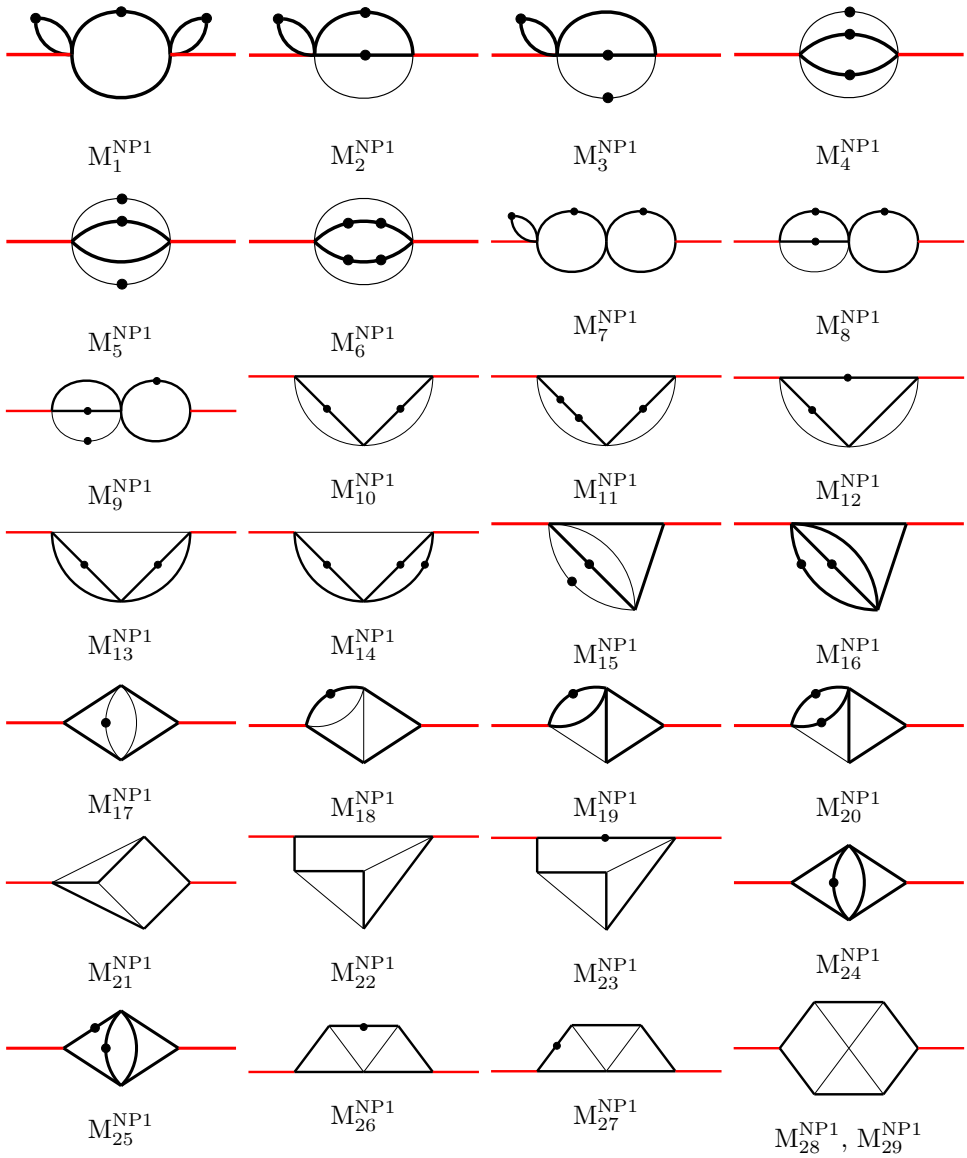

Figure 11. Master integrals in the NP1 topology. The thick black and red lines stand for the massive bottom quark and the Higgs boson, respectively. One black dot indicates one additional power of the corresponding propagator. The numerators can be found in the text.

A Topologies of the master integrals

The topology diagrams of the master integrals in the NP1 and P1 are displayed in figure 11 and figure 12.

B Asymptotic expansion of the master integrals

In the limit of $z \to \infty$ ($m_b \to 0$), all the master integrals that are needed in $X_{2,b\bar{b}b\bar{b}}^{y_b y_b}$ can be expanded with the method in [91]. Here we present the asymptotic expansion results:

$$F_1^{4b}(z)|_{z\to\infty} = 12\pi\log^2(z) - 4\pi^3 - \frac{1}{z}\left(144\pi\log^2(z) + 144\pi\log(z) - 48\pi^3\right) + \mathcal{O}(z^{-2}),$$

JHEP03(2024)068

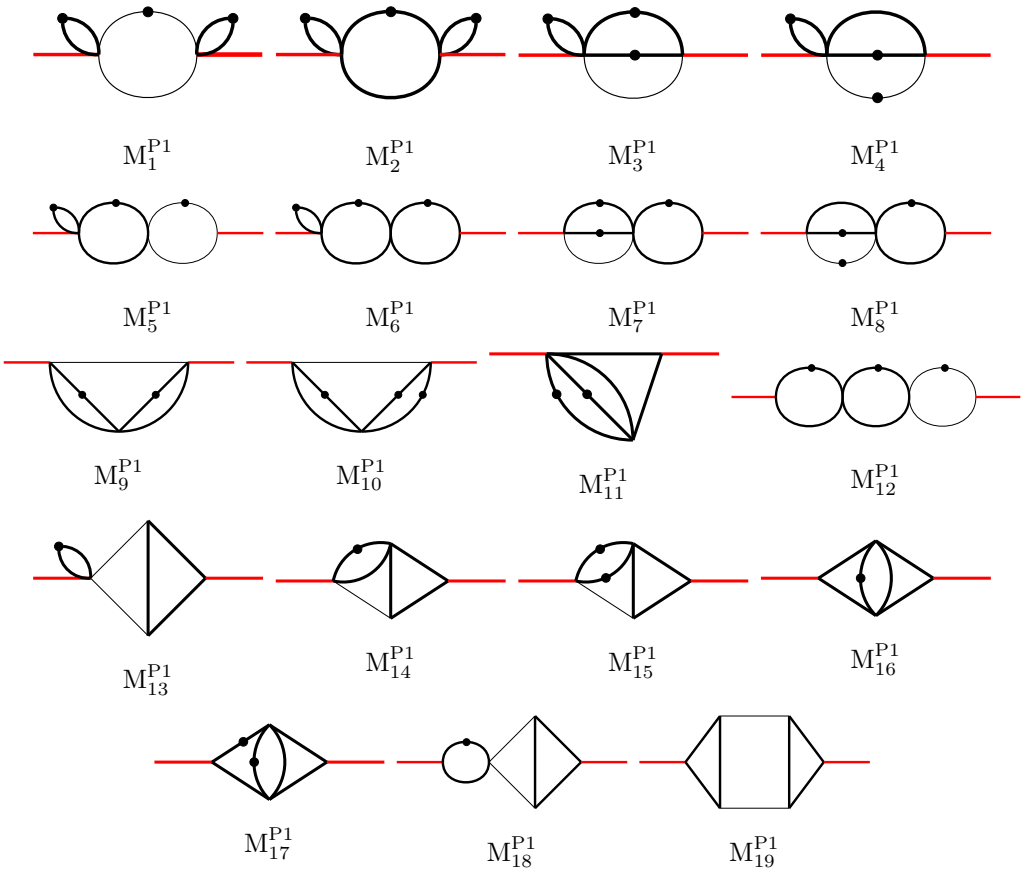

Figure 12. Master integrals in the P1 topology. The thick black and red lines stand for the massive bottom quark and the Higgs boson, respectively. One black dot indicates one additional power of the corresponding propagator.

$$
\begin{aligned}
F_2^{4b}(z)|_{z\to\infty} &= 18\pi\log^2(z)-24\pi\log(z)-6\pi^3 \\
&\quad -\frac{1}{z}\left(72\pi\log^2(z)+72\pi\log(z)-24\pi^3-144\pi\right)+\mathcal{O}(z^{-2}), \\
F_3^{4b}(z)|_{z\to\infty} &= -6\pi\log(z)+12\pi \\
&\quad +\frac{1}{z}\left(36\pi\log^2(z)-36\pi\log(z)-12\pi^3+36\pi\right)+\mathcal{O}(z^{-2}), \\
F_4^{4b}(z)|_{z\to\infty} &= 12\pi\log^2(z)-8\pi^3 \\
&\quad -\frac{1}{z}\left(72\pi\log^2(z)+48\pi\log(z)-24\pi^3+96\pi\right)+\mathcal{O}(z^{-2}), \\
F_5^{4b}(z)|_{z\to\infty} &= 4\pi\log^3(z)-4\pi^3\log(z)-96\pi\zeta(3) \\
&\quad +\frac{1}{z}\left(144\pi\log^2(z)+432\pi\log(z)-48\pi^3+432\pi\right)+\mathcal{O}(z^{-2}), \\
F_6^{4b}(z)|_{z\to\infty} &= -4\pi\log^3(z)+8\pi^3\log(z)-36\pi\zeta(3)-8\pi^3\log(2) \\
&\quad -\frac{1}{z}\left(24\pi\log^2(z)+96\pi\log(z)+8\pi^3\right)+\mathcal{O}(z^{-2}),
\end{aligned}
$$

JHEP03(2024)068

$$
\begin{aligned}
F_7^{4b}(z)|_{z\to\infty} &= 4\pi\log^3(z)-8\pi^3\log(z)+72\pi\zeta(3)\\
&\quad+\frac{1}{z}\Big(48\pi\log^2(z)+144\pi\log(z)-8\pi^3+48\pi\Big)+\mathcal{O}(z^{-2}),\\
F_8^{4b}(z)|_{z\to\infty} &= -2\pi\log^3(z)+6\pi^3\log(z)-60\pi\zeta(3)-8\pi^3\log(2)\\
&\quad-\frac{1}{z}\Big(12\pi\log^2(z)-4\pi^3-96\pi\Big)+\mathcal{O}(z^{-2}),\\
F_9^{4b}(z)|_{z\to\infty} &= -\frac{1}{2}\pi\log^4(z)+\pi^3\log^2(z)+24\pi\zeta(3)\log(z)-\frac{19\pi^5}{10}\\
&\quad+\frac{1}{z}\Big(4\pi\log^3(z)+24\pi\log^2(z)-4\pi^3\log(z)+144\pi\log(z)-48\pi\zeta(3)+8\pi^3+240\pi\Big)\\
&\quad+\mathcal{O}(z^{-2}),\\
F_{10}^{4b}(z)|_{z\to\infty} &= \pi\log^4(z)-4\pi^3\log^2(z)+72\pi\zeta(3)\log(z)+\frac{13\pi^5}{15}\\
&\quad-\frac{1}{z}\Big(48\pi\log^2(z)+240\pi\log(z)-8\pi^3+288\pi\Big)+\mathcal{O}(z^{-2}),\\
F_{12}^{4b}(z)|_{z\to\infty} &= \frac{3}{4}\pi\log^4(z)-\frac{7}{2}\pi^3\log^2(z)+96\pi\zeta(3)\log(z)-\frac{43\pi^5}{60}\\
&\quad-\frac{1}{z}\Big(6\pi\log^3(z)+30\pi\log^2(z)-14\pi^3\log(z)+96\pi\log(z)+192\pi\zeta(3)-10\pi^3+36\pi\Big)\\
&\quad+\mathcal{O}(z^{-2}).
\end{aligned}
\tag{B.1}
$$

Open Access. This article is distributed under the terms of the Creative Commons Attribution License (CC-BY4.0), which permits any use, distribution and reproduction in any medium, provided the original author(s) and source are credited.

References

[1] ATLAS collaboration, *A detailed map of Higgs boson interactions by the ATLAS experiment ten years after the discovery*, *Nature* **607** (2022) 52 [*Erratum ibid.* **612** (2022) E24] [arXiv:2207.00092] [INSPIRE].

[2] CMS collaboration, *A portrait of the Higgs boson by the CMS experiment ten years after the discovery*, *Nature* **607** (2022) 60 [arXiv:2207.00043] [INSPIRE].

[3] Particle Data Group collaboration, *Review of Particle Physics*, *PTEP* **2022** (2022) 083C01 [INSPIRE].

[4] CMS collaboration, *Measurement of the top quark Yukawa coupling from $t\bar{t}$ kinematic distributions in the lepton+jets final state in proton-proton collisions at $\sqrt{s} = 13$ TeV*, *Phys. Rev. D* **100** (2019) 072007 [arXiv:1907.01590] [INSPIRE].

[5] CMS collaboration, *Measurement of the top quark Yukawa coupling from $t\bar{t}$ kinematic distributions in the dilepton final state in proton-proton collisions at $\sqrt{s} = 13$ TeV*, *Phys. Rev. D* **102** (2020) 092013 [arXiv:2009.07123] [INSPIRE].

[6] ATLAS collaboration, *Observation of $H \to b\bar{b}$ decays and VH production with the ATLAS detector*, *Phys. Lett. B* **786** (2018) 59 [arXiv:1808.08238] [INSPIRE].

[7] CMS collaboration, *Observation of Higgs boson decay to bottom quarks*, *Phys. Rev. Lett.* **121** (2018) 121801 [arXiv:1808.08242] [INSPIRE].

JHEP03(2024)068

[8] ATLAS collaboration, *Direct constraint on the Higgs-charm coupling from a search for Higgs boson decays into charm quarks with the ATLAS detector*, Eur. Phys. J. C **82** (2022) 717 [arXiv:2201.11428] [INSPIRE].

[9] CMS collaboration, *Search for Higgs Boson Decay to a Charm Quark-Antiquark Pair in Proton-Proton Collisions at* $\sqrt{s} = 13$ *TeV*, Phys. Rev. Lett. **131** (2023) 061801 [arXiv:2205.05550] [INSPIRE].

[10] ATLAS collaboration, *Measurements of Higgs boson production cross-sections in the* $H \to \tau^+\tau^-$ *decay channel in pp collisions at* $\sqrt{s} = 13$ *TeV with the ATLAS detector*, JHEP **08** (2022) 175 [arXiv:2201.08269] [INSPIRE].

[11] CMS collaboration, *Measurements of Higgs boson production in the decay channel with a pair of* τ *leptons in proton-proton collisions at* $\sqrt{s} = 13$ *TeV*, Eur. Phys. J. C **83** (2023) 562 [arXiv:2204.12957] [INSPIRE].

[12] ATLAS collaboration, *A search for the dimuon decay of the Standard Model Higgs boson with the ATLAS detector*, Phys. Lett. B **812** (2021) 135980 [arXiv:2007.07830] [INSPIRE].

[13] CMS collaboration, *Evidence for Higgs boson decay to a pair of muons*, JHEP **01** (2021) 148 [arXiv:2009.04363] [INSPIRE].

[14] F. An et al., *Precision Higgs physics at the CEPC*, Chin. Phys. C **43** (2019) 043002 [arXiv:1810.09037] [INSPIRE].

[15] CEPC Physics Study Group collaboration, *The physics potential of the CEPC. Prepared for the US Snowmass Community Planning Exercise (Snowmass 2021)*, in the proceedings of the *Snowmass 2021*, Seattle, U.S.A., July 17–26 (2022) [arXiv:2205.08553] [INSPIRE].

[16] Y. Zhu, H. Cui and M. Ruan, *The Higgs*$\to b\bar{b}, c\bar{c}$*, gg measurement at CEPC*, JHEP **11** (2022) 100 [arXiv:2203.01469] [INSPIRE].

[17] E. Braaten and J.P. Leveille, *Higgs Boson Decay and the Running Mass*, Phys. Rev. D **22** (1980) 715 [INSPIRE].

[18] N. Sakai, *Perturbative QCD Corrections to the Hadronic Decay Width of the Higgs Boson*, Phys. Rev. D **22** (1980) 2220 [INSPIRE].

[19] P. Janot, *First Order QED and QCD Radiative Corrections to Higgs Decay Into Massive Fermions*, Phys. Lett. B **223** (1989) 110 [INSPIRE].

[20] M. Drees and K.-I. Hikasa, *Note on qcd corrections to hadronic higgs decay*, Phys. Lett. B **240** (1990) 455 [*Erratum ibid.* **262** (1991) 497] [INSPIRE].

[21] A. Dabelstein and W. Hollik, *Electroweak corrections to the fermionic decay width of the standard Higgs boson*, Z. Phys. C **53** (1992) 507 [INSPIRE].

[22] B.A. Kniehl, *Radiative corrections for* $H \to f\bar{f}(\gamma)$ *in the standard model*, Nucl. Phys. B **376** (1992) 3 [INSPIRE].

[23] A.L. Kataev, *The Order* $O(\bar{\alpha}al\bar{p}ha_s)$ *and* $O(al\bar{p}ha^2)$ *corrections to the decay width of the neutral Higgs boson to the anti-b b pair*, JETP Lett. **66** (1997) 327 [hep-ph/9708292] [INSPIRE].

[24] L. Mihaila, B. Schmidt and M. Steinhauser, $\Gamma(H \to b\bar{b})$ *to order* $\alpha\alpha_s$, Phys. Lett. B **751** (2015) 442 [arXiv:1509.02294] [INSPIRE].

[25] S.G. Gorishnii, A.L. Kataev, S.A. Larin and L.R. Surguladze, *Corrected Three Loop QCD Correction to the Correlator of the Quark Scalar Currents and* $\Gamma_{\rm tot}(H^0 \to$ *Hadrons*$)$, Mod. Phys. Lett. A **5** (1990) 2703 [INSPIRE].

JHEP03(2024)068

[26] K.G. Chetyrkin, *Correlator of the quark scalar currents and $\Gamma_{\rm tot}(H \to hadrons)$ at $O(\alpha_s^3)$ in pQCD*, *Phys. Lett. B* **390** (1997) 309 [hep-ph/9608318] [INSPIRE].

[27] P.A. Baikov, K.G. Chetyrkin and J.H. Kuhn, *Scalar correlator at $O(\alpha_s^4)$, Higgs decay into b-quarks and bounds on the light quark masses*, *Phys. Rev. Lett.* **96** (2006) 012003 [hep-ph/0511063] [INSPIRE].

[28] F. Herzog et al., *On Higgs decays to hadrons and the R-ratio at N^4LO*, *JHEP* **08** (2017) 113 [arXiv:1707.01044] [INSPIRE].

[29] X. Chen, P. Jakubčík, M. Marcoli and G. Stagnitto, *The parton-level structure of Higgs decays to hadrons at N^3LO*, *JHEP* **06** (2023) 185 [arXiv:2304.11180] [INSPIRE].

[30] C. Anastasiou, F. Herzog and A. Lazopoulos, *The fully differential decay rate of a Higgs boson to bottom-quarks at NNLO in QCD*, *JHEP* **03** (2012) 035 [arXiv:1110.2368] [INSPIRE].

[31] V. Del Duca et al., *Higgs boson decay into b-quarks at NNLO accuracy*, *JHEP* **04** (2015) 036 [arXiv:1501.07226] [INSPIRE].

[32] R. Mondini, M. Schiavi and C. Williams, *N^3LO predictions for the decay of the Higgs boson to bottom quarks*, *JHEP* **06** (2019) 079 [arXiv:1904.08960] [INSPIRE].

[33] W. Bernreuther, L. Chen and Z.-G. Si, *Differential decay rates of CP-even and CP-odd Higgs bosons to top and bottom quarks at NNLO QCD*, *JHEP* **07** (2018) 159 [arXiv:1805.06658] [INSPIRE].

[34] A. Behring and W. Bizoń, *Higgs decay into massive b-quarks at NNLO QCD in the nested soft-collinear subtraction scheme*, *JHEP* **01** (2020) 189 [arXiv:1911.11524] [INSPIRE].

[35] G. Somogyi and F. Tramontano, *Fully exclusive heavy quark-antiquark pair production from a colourless initial state at NNLO in QCD*, *JHEP* **11** (2020) 142 [arXiv:2007.15015] [INSPIRE].

[36] W. Bizoń, E. Re and G. Zanderighi, *NNLOPS description of the $H \to b\bar{b}$ decay with MiNLO*, *JHEP* **06** (2020) 006 [arXiv:1912.09982] [INSPIRE].

[37] Y.L. Hu, C.L. Sun, X.-M. Shen and J. Gao, *Hadronic decays of Higgs boson at NNLO matched with parton shower*, *JHEP* **08** (2021) 122 [arXiv:2101.08916] [INSPIRE].

[38] R. Mondini and C. Williams, *$H \to b\bar{b}j$ at next-to-next-to-leading order accuracy*, *JHEP* **06** (2019) 120 [arXiv:1904.08961] [INSPIRE].

[39] J. Gao, *Higgs boson decay into four bottom quarks in the SM and beyond*, *JHEP* **08** (2019) 174 [arXiv:1905.04865] [INSPIRE].

[40] A. Primo, G. Sasso, G. Somogyi and F. Tramontano, *Exact Top Yukawa corrections to Higgs boson decay into bottom quarks*, *Phys. Rev. D* **99** (2019) 054013 [arXiv:1812.07811] [INSPIRE].

[41] R. Mondini, U. Schubert and C. Williams, *Top-induced contributions to $H \to b\bar{b}$ and $H \to c\bar{c}$ at $\mathcal{O}(\alpha_s^3)$*, *JHEP* **12** (2020) 058 [arXiv:2006.03563] [INSPIRE].

[42] K.G. Chetyrkin and M. Steinhauser, *Complete QCD corrections of order $O(\alpha_s^3)$ to the hadronic Higgs decay*, *Phys. Lett. B* **408** (1997) 320 [hep-ph/9706462] [INSPIRE].

[43] J. Davies, M. Steinhauser and D. Wellmann, *Completing the hadronic Higgs boson decay at order α_s^4*, *Nucl. Phys. B* **920** (2017) 20 [arXiv:1703.02988] [INSPIRE].

[44] K.G. Chetyrkin and A. Kwiatkowski, *Second order QCD corrections to scalar and pseudoscalar Higgs decays into massive bottom quarks*, *Nucl. Phys. B* **461** (1996) 3 [hep-ph/9505358] [INSPIRE].

JHEP03(2024)068

[45] R. Harlander and M. Steinhauser, *Higgs decay to top quarks at $O(\alpha_s^2)$*, Phys. Rev. D **56** (1997) 3980 [hep-ph/9704436] [INSPIRE].

[46] T. Hahn, *Generating Feynman diagrams and amplitudes with FeynArts 3*, Comput. Phys. Commun. **140** (2001) 418 [hep-ph/0012260] [INSPIRE].

[47] V. Shtabovenko, R. Mertig and F. Orellana, *FeynCalc 9.3: New features and improvements*, Comput. Phys. Commun. **256** (2020) 107478 [arXiv:2001.04407] [INSPIRE].

[48] F.V. Tkachov, *A theorem on analytical calculability of 4-loop renormalization group functions*, Phys. Lett. B **100** (1981) 65 [INSPIRE].

[49] K.G. Chetyrkin and F.V. Tkachov, *Integration by parts: The algorithm to calculate β-functions in 4 loops*, Nucl. Phys. B **192** (1981) 159 [INSPIRE].

[50] S. Laporta, *High-precision calculation of multiloop Feynman integrals by difference equations*, Int. J. Mod. Phys. A **15** (2000) 5087 [hep-ph/0102033] [INSPIRE].

[51] A.V. Smirnov and F.S. Chuharev, *FIRE6: Feynman Integral REduction with Modular Arithmetic*, Comput. Phys. Commun. **247** (2020) 106877 [arXiv:1901.07808] [INSPIRE].

[52] J. Klappert, F. Lange, P. Maierhöfer and J. Usovitsch, *Integral reduction with Kira 2.0 and finite field methods*, Comput. Phys. Commun. **266** (2021) 108024 [arXiv:2008.06494] [INSPIRE].

[53] R.N. Lee and A.I. Onishchenko, *ϵ-regular basis for non-polylogarithmic multiloop integrals and total cross section of the process $e^+e^- \to 2(Q\bar{Q})$*, JHEP **12** (2019) 084 [arXiv:1909.07710] [INSPIRE].

[54] A.V. Kotikov, *Differential equations method: New technique for massive Feynman diagrams calculation*, Phys. Lett. B **254** (1991) 158 [INSPIRE].

[55] A.V. Kotikov, *Differential equation method: The calculation of N point Feynman diagrams*, Phys. Lett. B **267** (1991) 123 [INSPIRE].

[56] J.M. Henn, *Multiloop integrals in dimensional regularization made simple*, Phys. Rev. Lett. **110** (2013) 251601 [arXiv:1304.1806] [INSPIRE].

[57] A.B. Goncharov, *Multiple polylogarithms, cyclotomy and modular complexes*, Math. Res. Lett. **5** (1998) 497 [arXiv:1105.2076] [INSPIRE].

[58] M. Argeri et al., *Magnus and Dyson Series for Master Integrals*, JHEP **03** (2014) 082 [arXiv:1401.2979] [INSPIRE].

[59] R.N. Lee and A.I. Onishchenko, *Master integrals for bipartite cuts of three-loop photon self energy*, JHEP **04** (2021) 177 [arXiv:2012.04230].

[60] L.-B. Chen and J. Wang, *Analytic two-loop master integrals for tW production at hadron colliders: I*, Chin. Phys. C **45** (2021) 123106 [arXiv:2106.12093] [INSPIRE].

[61] C. Duhr and F. Dulat, *PolyLogTools — polylogs for the masses*, JHEP **08** (2019) 135 [arXiv:1904.07279] [INSPIRE].

[62] J. Vollinga and S. Weinzierl, *Numerical evaluation of multiple polylogarithms*, Comput. Phys. Commun. **167** (2005) 177 [hep-ph/0410259] [INSPIRE].

[63] C.W. Bauer, A. Frink and R. Kreckel, *Introduction to the GiNaC framework for symbolic computation within the C++ programming language*, J. Symb. Comput. **33** (2002) 1 [cs/0004015] [INSPIRE].

[64] X. Liu, Y.-Q. Ma and C.-Y. Wang, *A Systematic and Efficient Method to Compute Multi-loop Master Integrals*, Phys. Lett. B **779** (2018) 353 [arXiv:1711.09572] [INSPIRE].

JHEP03(2024)068

[65] X. Liu, Y.-Q. Ma, W. Tao and P. Zhang, *Calculation of Feynman loop integration and phase-space integration via auxiliary mass flow*, Chin. Phys. C **45** (2021) 013115 [arXiv:2009.07987] [INSPIRE].

[66] X. Liu and Y.-Q. Ma, *AMFlow: A Mathematica package for Feynman integrals computation via auxiliary mass flow*, Comput. Phys. Commun. **283** (2023) 108565 [arXiv:2201.11669] [INSPIRE].

[67] Z.-F. Liu and Y.-Q. Ma, *Determining Feynman Integrals with Only Input from Linear Algebra*, Phys. Rev. Lett. **129** (2022) 222001 [arXiv:2201.11637] [INSPIRE].

[68] H. Ferguson and D. Bailey, *A polynomial time, numerically stable integer relation algorithm*, RNR-91-032 (1992).

[69] H.R.P. Ferguson, D.H. Bailey and S. Arno, *Analysis of PSLQ, an integer relation finding algorithm*, Math. Comput. **68** (1999) 351.

[70] W.E. Caswell, *Asymptotic Behavior of Nonabelian Gauge Theories to Two Loop Order*, Phys. Rev. Lett. **33** (1974) 244 [INSPIRE].

[71] D.R.T. Jones, *Two Loop Diagrams in Yang-Mills Theory*, Nucl. Phys. B **75** (1974) 531 [INSPIRE].

[72] M. Czakon, A. Mitov and S. Moch, *Heavy-quark production in gluon fusion at two loops in QCD*, Nucl. Phys. B **798** (2008) 210 [arXiv:0707.4139] [INSPIRE].

[73] K. Melnikov and T. Ritbergen, *The three loop relation between the MS-bar and the pole quark masses*, Phys. Lett. B **482** (2000) 99 [hep-ph/9912391] [INSPIRE].

[74] D.J. Broadhurst, N. Gray and K. Schilcher, *Gauge invariant on-shell Z_2 in QED, QCD and the effective field theory of a static quark*, Z. Phys. C **52** (1991) 111 [INSPIRE].

[75] N. Gray, D.J. Broadhurst, W. Grafe and K. Schilcher, *Three Loop Relation of Quark (Modified) Ms and Pole Masses*, Z. Phys. C **48** (1990) 673 [INSPIRE].

[76] P. Marquard, A.V. Smirnov, V.A. Smirnov and M. Steinhauser, *Quark Mass Relations to Four-Loop Order in Perturbative QCD*, Phys. Rev. Lett. **114** (2015) 142002 [arXiv:1502.01030] [INSPIRE].

[77] P. Marquard et al., *$\overline{\text{MS}}$-on-shell quark mass relation up to four loops in QCD and a general* SU(N) *gauge group*, Phys. Rev. D **94** (2016) 074025 [arXiv:1606.06754] [INSPIRE].

[78] T. Kinoshita, *Mass singularities of Feynman amplitudes*, J. Math. Phys. **3** (1962) 650 [INSPIRE].

[79] T.D. Lee and M. Nauenberg, *Degenerate Systems and Mass Singularities*, Phys. Rev. **133** (1964) B1549 [INSPIRE].

[80] J.M. Henn, A.V. Smirnov and V.A. Smirnov, *Evaluating Multiple Polylogarithm Values at Sixth Roots of Unity up to Weight Six*, Nucl. Phys. B **919** (2017) 315 [arXiv:1512.08389] [INSPIRE].

[81] H. Frellesvig, D. Tommasini and C. Wever, *On the reduction of generalized polylogarithms to Li_n and $Li_{2,2}$ and on the evaluation thereof*, JHEP **03** (2016) 189 [arXiv:1601.02649] [INSPIRE].

[82] J. Ablinger et al., *Heavy quark form factors at two loops*, Phys. Rev. D **97** (2018) 094022 [arXiv:1712.09889] [INSPIRE].

[83] M. Beneke, A. Signer and V.A. Smirnov, *Top quark production near threshold and the top quark mass*, Phys. Lett. B **454** (1999) 137 [hep-ph/9903260] [INSPIRE].

[84] A. Czarnecki and K. Melnikov, *Two loop QCD corrections to the heavy quark pair production cross-section in e^+e^- annihilation near the threshold*, Phys. Rev. Lett. **80** (1998) 2531 [hep-ph/9712222] [INSPIRE].

JHEP03(2024)068

[85] I.L. Solovtsov, O.P. Solovtsova and Y.D. Chernichenko, *Relativistic resummation of threshold singularities in the quasi-potential approach*, *Phys. Part. Nucl. Lett.* **2** (2005) 199 [INSPIRE].

[86] M. Beneke, J. Piclum and T. Rauh, *P-wave contribution to third-order top-quark pair production near threshold*, *Nucl. Phys. B* **880** (2014) 414 [arXiv:1312.4792] [INSPIRE].

[87] K.G. Chetyrkin, J.H. Kuhn and M. Steinhauser, *RunDec: A Mathematica package for running and decoupling of the strong coupling and quark masses*, *Comput. Phys. Commun.* **133** (2000) 43 [hep-ph/0004189] [INSPIRE].

[88] F. Herren and M. Steinhauser, *Version 3 of RunDec and CRunDec*, *Comput. Phys. Commun.* **224** (2018) 333 [arXiv:1703.03751] [INSPIRE].

[89] S.A. Larin, T. van Ritbergen and J.A.M. Vermaseren, *The large top quark mass expansion for Higgs boson decays into bottom quarks and into gluons*, *Phys. Lett. B* **362** (1995) 134 [hep-ph/9506465] [INSPIRE].

[90] J. Ellis, *TikZ-Feynman: Feynman diagrams with TikZ*, *Comput. Phys. Commun.* **210** (2017) 103 [arXiv:1601.05437] [INSPIRE].

[91] R.N. Lee, A.A. Lyubyakin and V.A. Stotsky, *Total cross sections of $e\gamma \to eX\bar{X}$ processes with $X = \mu, \gamma, e$ via multiloop methods*, *JHEP* **01** (2021) 144 [arXiv:2010.15430] [INSPIRE].

JHEP03(2024)068